

Chapter 4. Complexes of π -bound Ligands

Outline

A. Alkene complexes

B. Butadiene complexes

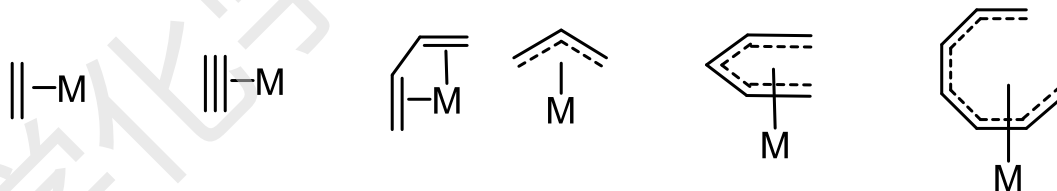
C. Alkyne complexes

D. Enyl Complexes

E. Complexes of the cyclic π -perimeters C_nH_n

F. Metal π -complexes of Unsaturated ligands with heteroatoms

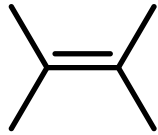
G. Characterization of organometallic compounds



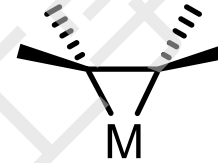
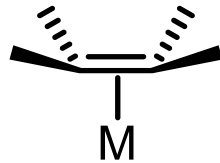


A. Alkene complexes

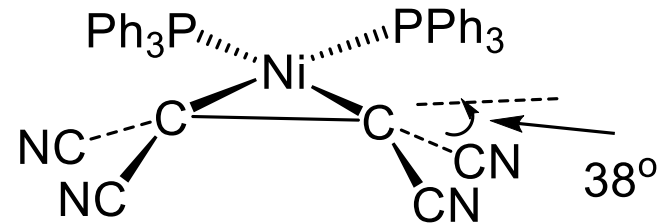
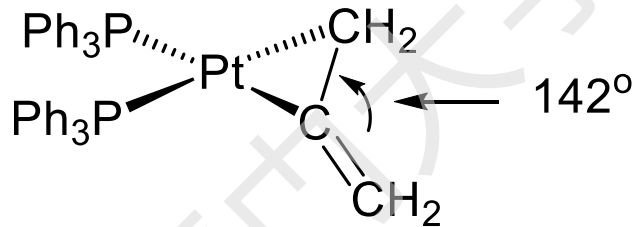
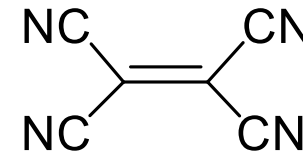
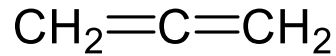
1. Structure and bonding of monoolefin complexes



planar

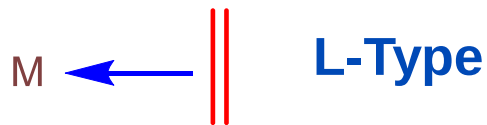


Non-planar





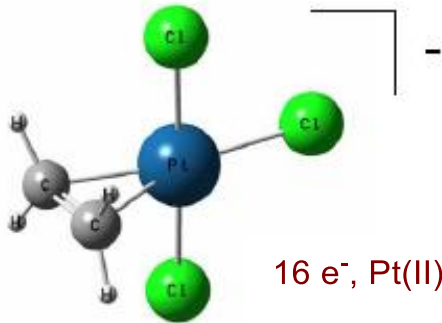
Types of Alkene complexes



π -complex



metallacyclopentane



16 e^- , Pt(II) $4d^8$

r_{C-C} [Å]

H-C-H angle($^\circ$)

C-C-H angle($^\circ$)

free C_2H_4	C_2H_6
1.335 (3)	1.532 (2)

116.6

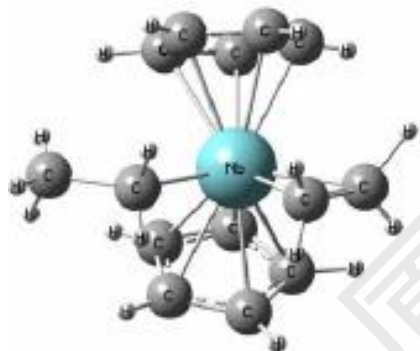
107.4

$K[PtCl_3(C_2H_4)]$

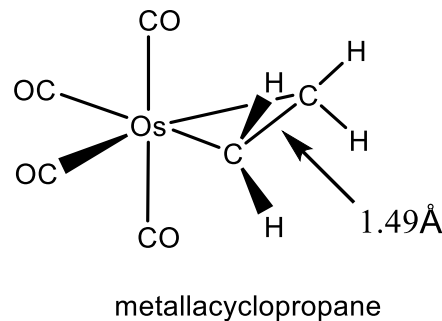
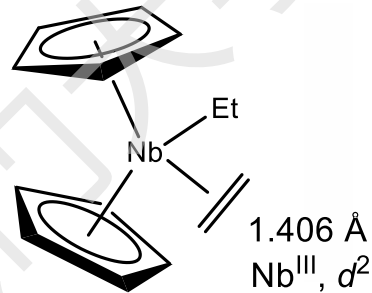
1.375 (4)

114.9

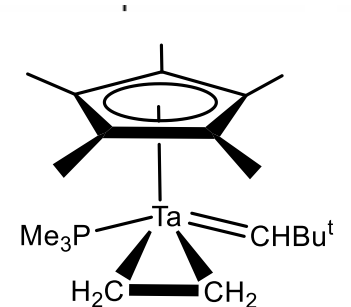
121.1



18 e^- Nb(III) - 4d metal



18 e^- , Os(II) - 5d metal

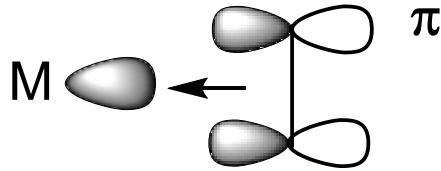
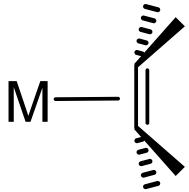


18 e^- , Ta(V) - 5d metal



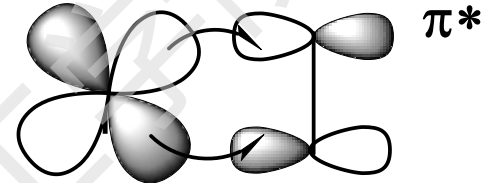
Bonding of alkene complexes

General bonding schemes



1. e^- from $\pi(\text{olefin}) \rightarrow \sigma(\text{M})$

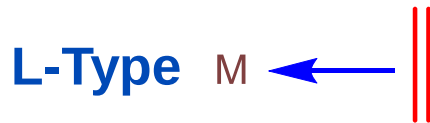
σ -donation



2. e^- from $d\pi(\text{M}) \rightarrow \pi^*(\text{olefin})$

π -backdonation

Bonding and type of alkene complexes



Favored if σ -donation is dominant

- electron-deficient metal center
- alkenes with electron donating group



Favored if π -backdonation is dominant

- electron-rich metal center
- alkenes with electron withdrawing group

L-type alkenes are electrophilic, X-type alkenes are nucleophilic

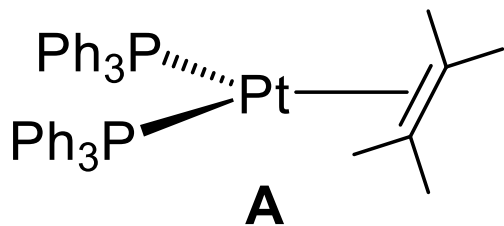
Factors favoring X₂-type binding

- strong donor ligands
- a net negative charge on the complex
- low-oxidation state metals



Bonding of alkene complexes

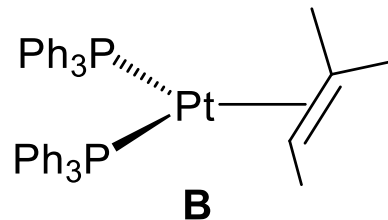
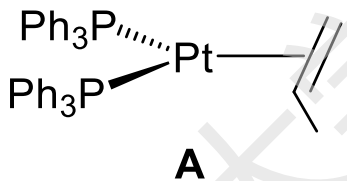
Q1. Consider the complexes A below. Which of the following olefins would you expect to have the strongest Pt- olefin bond?



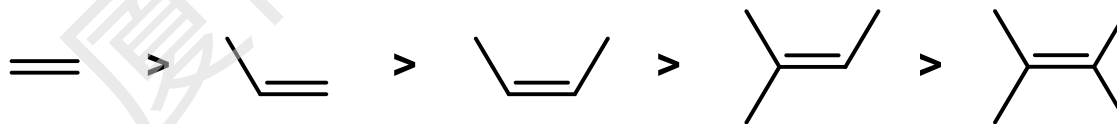
(a) $\text{CH}_2=\text{CH}_2$ (b) $\text{CH}_3-\text{CH}=\text{CH}_2$ (c) $\text{CH}_2=\text{CHCN}$

For low valent metals, e.g. Pt(0), Ni(0), Pd(0), backbonding is important. Electron-deficient olefin forms the more stable olefin complexes.

Q2. Which of the following olefin complexes have a stronger Pt- olefin bond?



Usual binding strength of olefins:



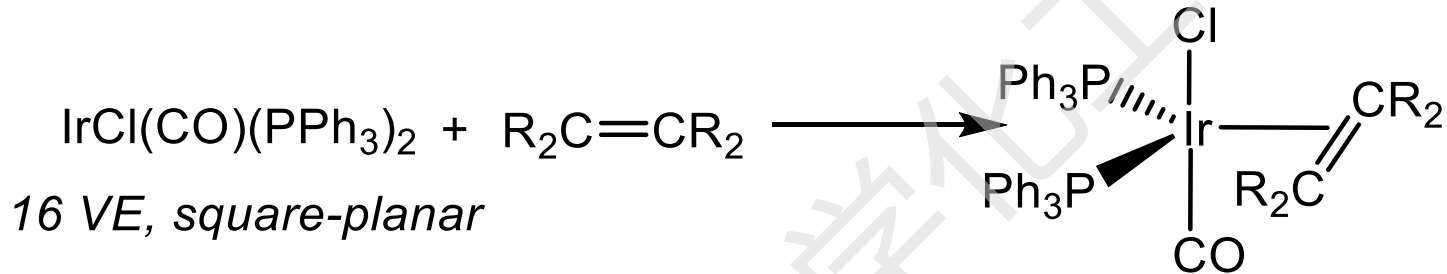
Affinity of alkenes are affected by both electronic (bonding) and steric effect



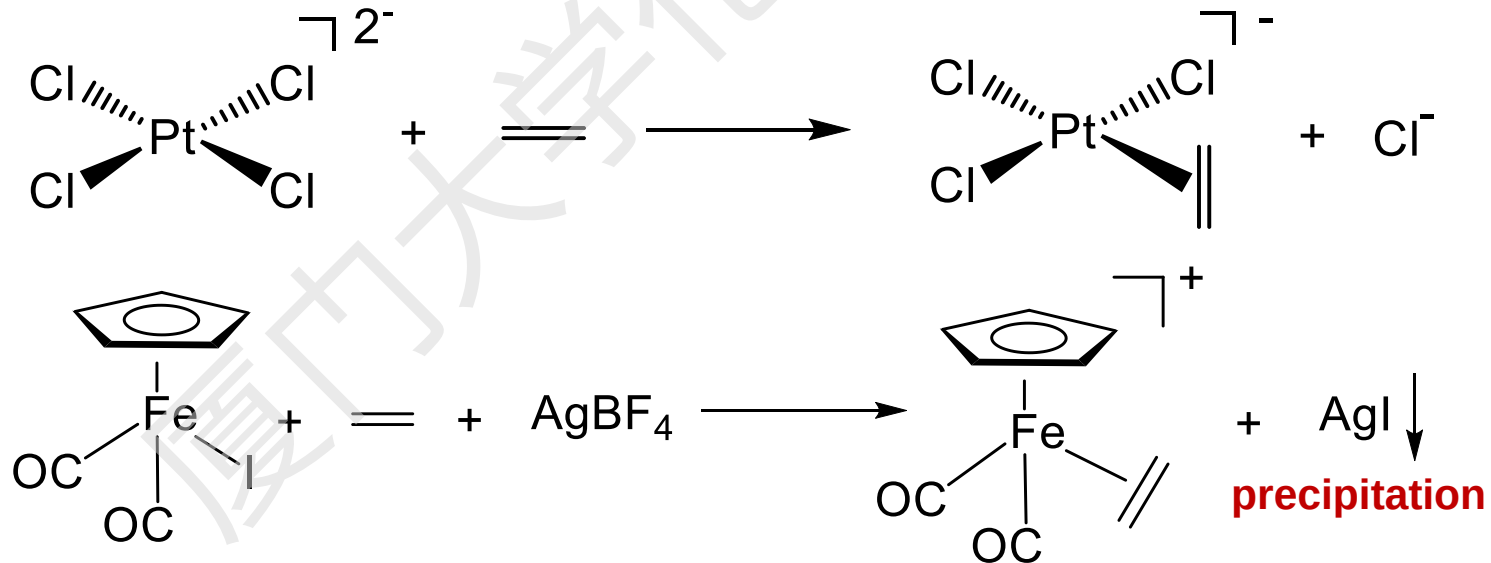
2. Common pathways to olefin complexes

A variety of alkene complexes are known. Common starting material include alkenes and metal alkyl, allyl complexes.

a. By addition of olefin to unsaturated complexes.



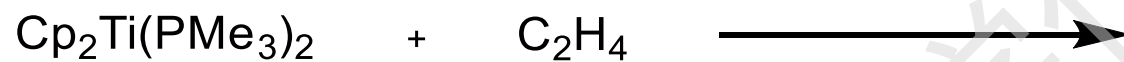
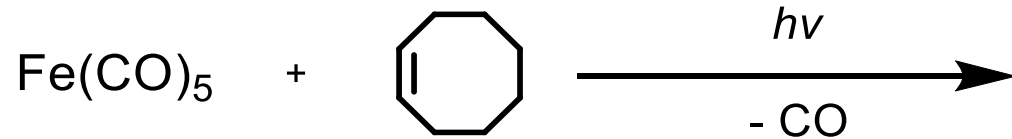
b. From Substitution reactions: $\text{L}_n\text{M-L}' + \text{olefin} \longrightarrow \text{L}' + \text{L}_n\text{M}(\text{olefin})$



TI^+ salts (e.g. TIBF_4) are also popular reagents for removing halide (TIX, insoluble).



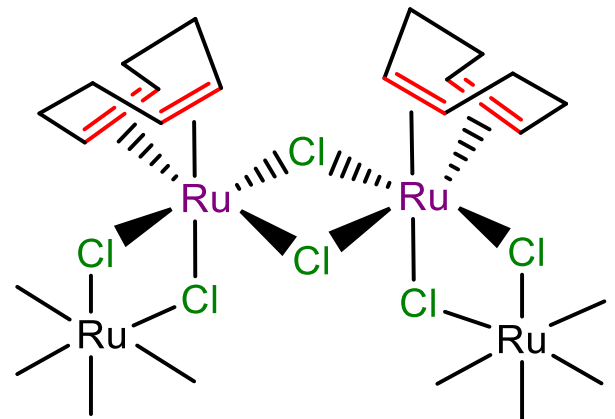
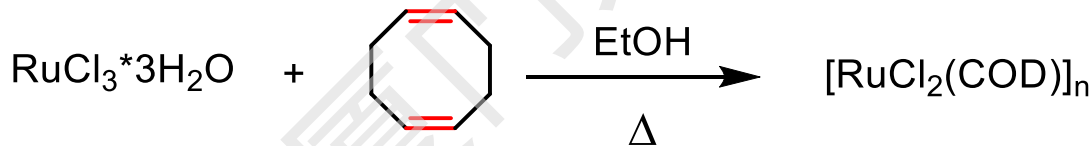
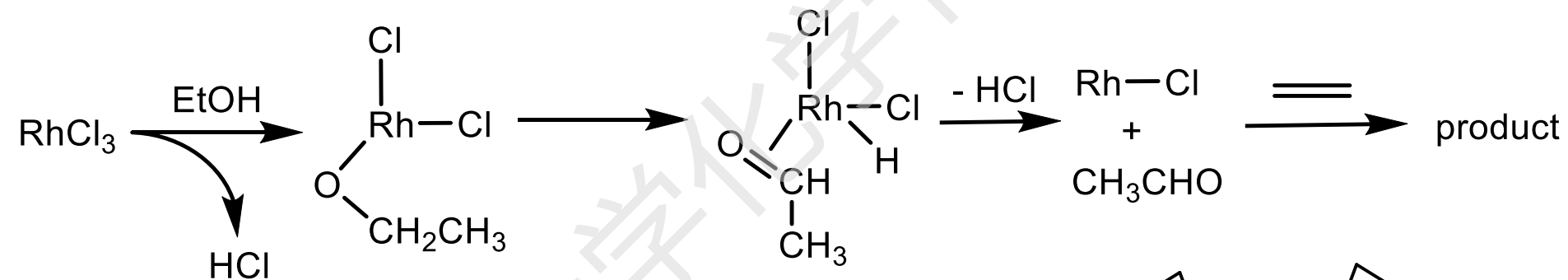
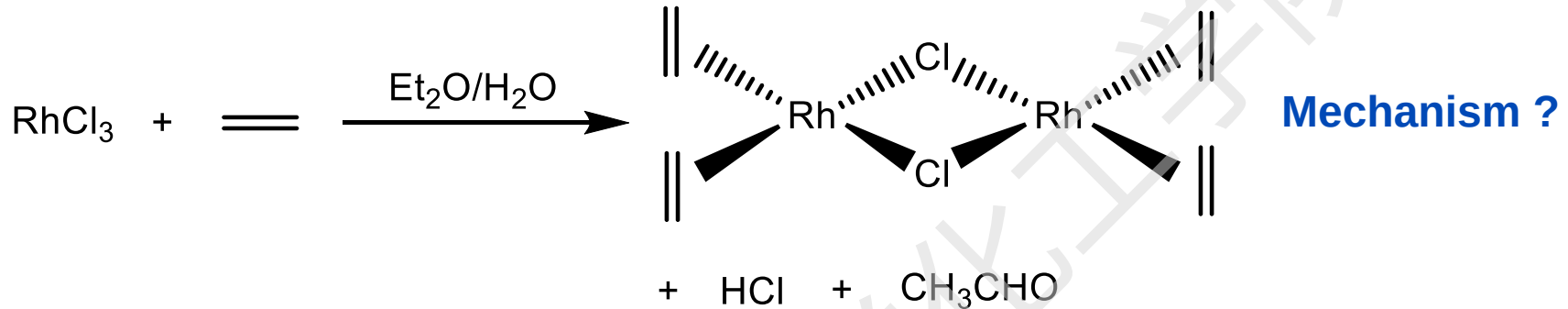
Practice Problems





2. Common pathways to olefin complexes

c. Metal salt + olefin + reducing agents. e.g.

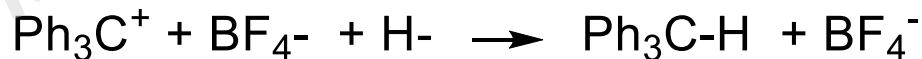
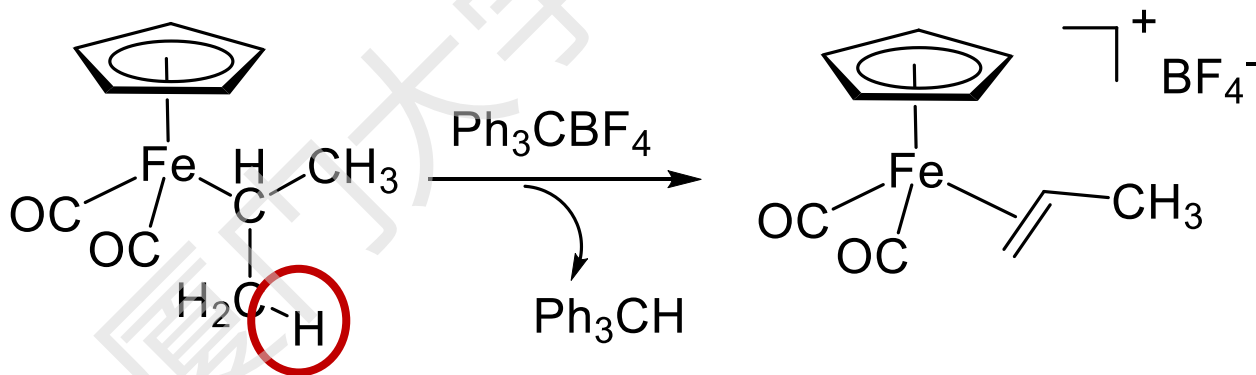
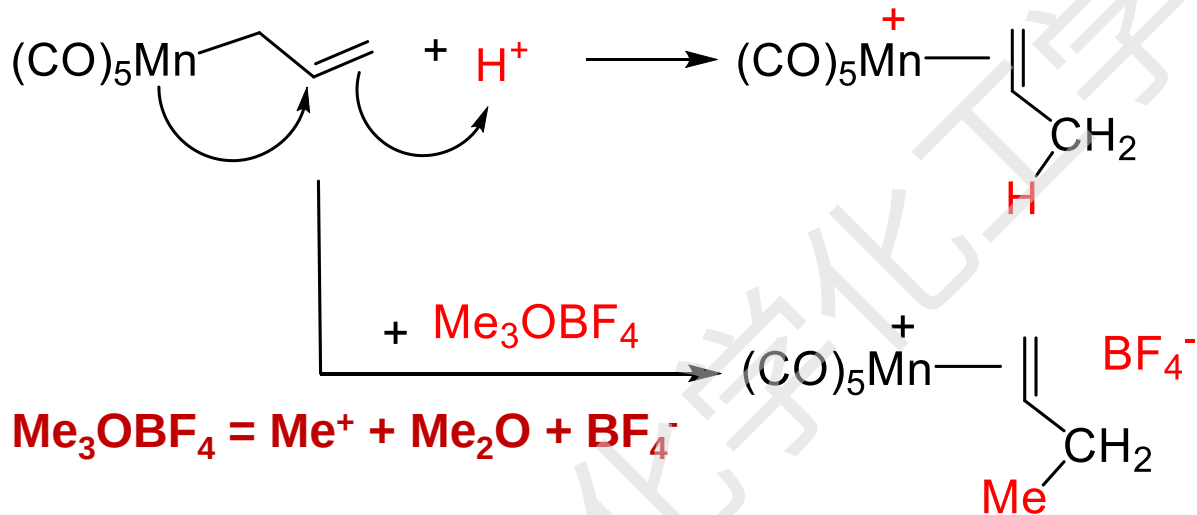


18e⁻, Ru(II), d⁶



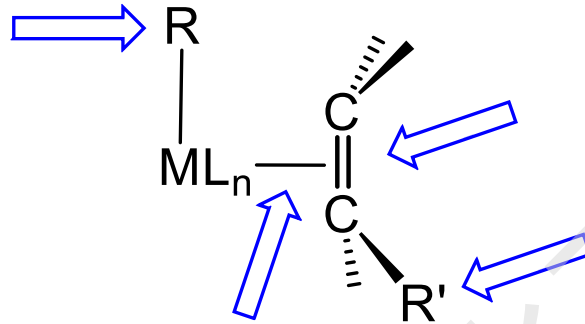
2. Common pathways to olefin complexes

d. From alkyl and related species:

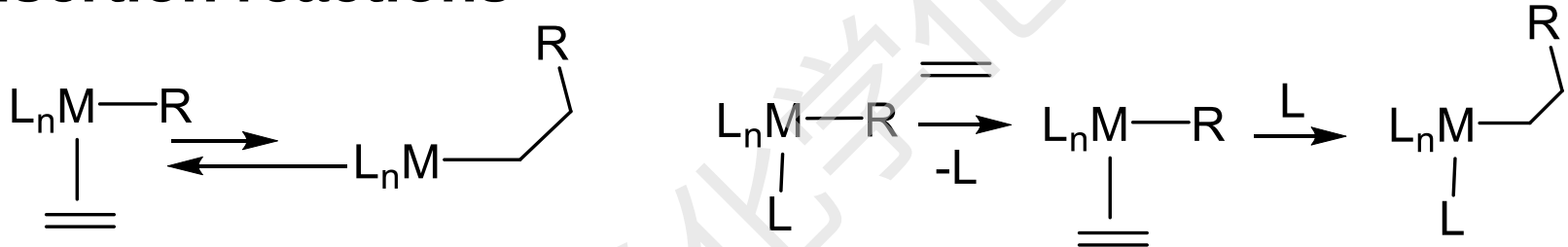




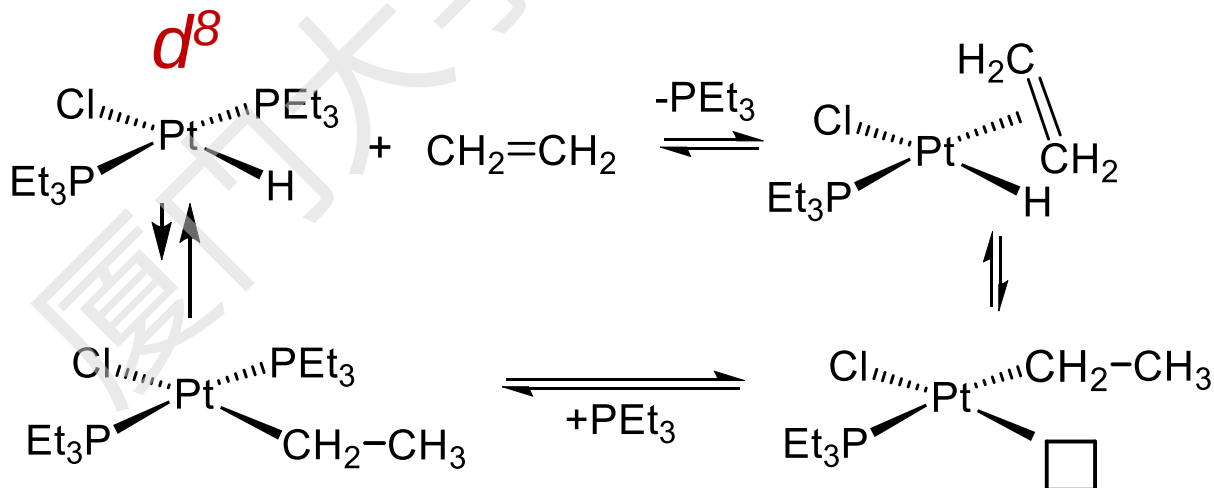
3. Reactivity of olefin complexes



a. Insertion reactions

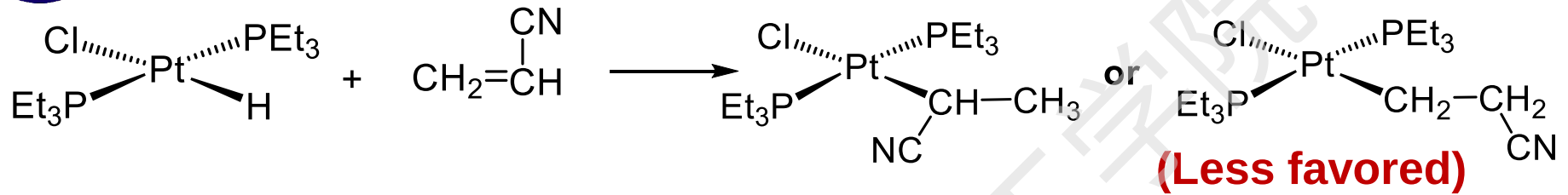
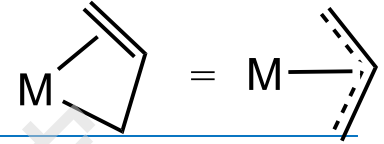


The reaction can be reversible, especially when $R = H$.





Practice Problems

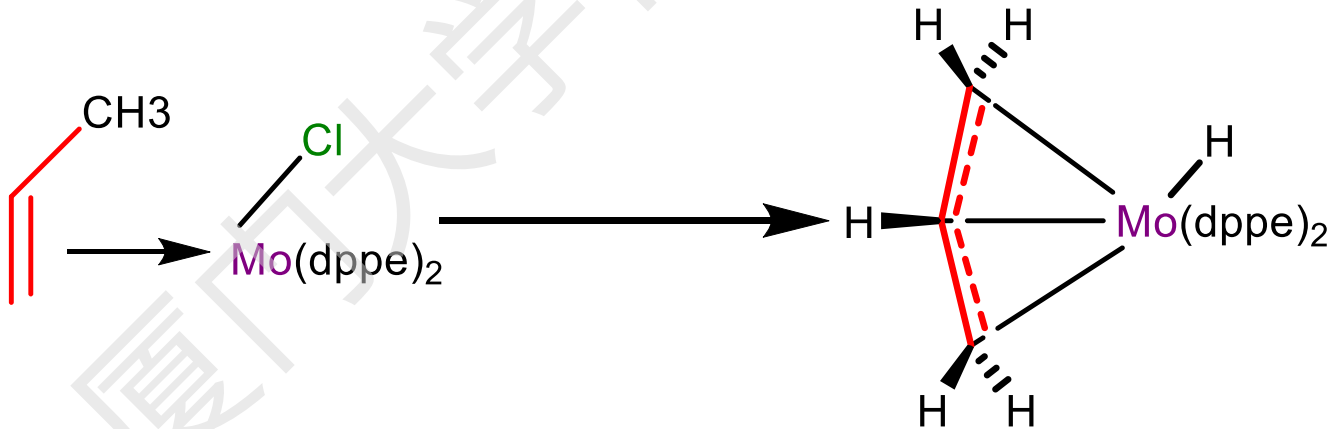
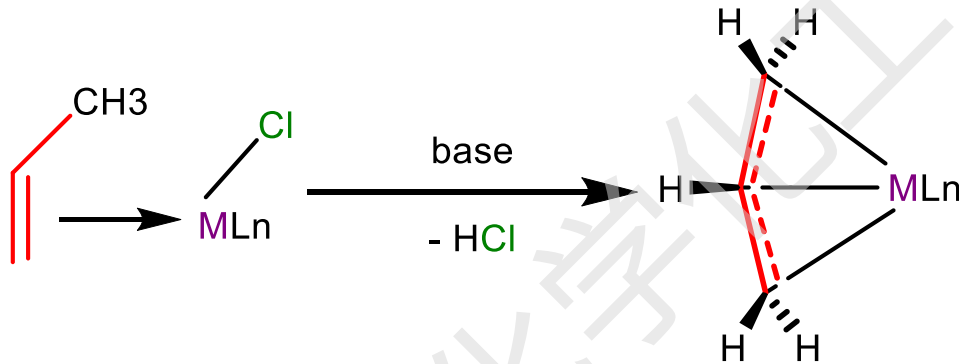
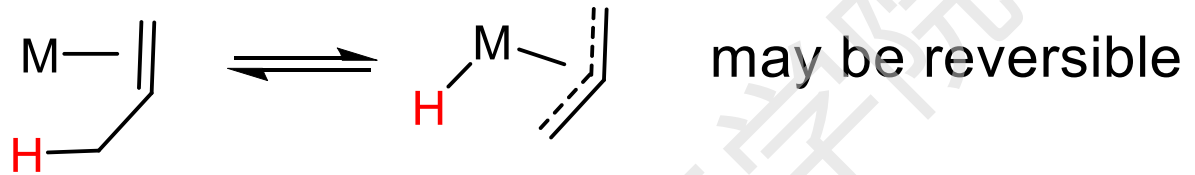


Reversible insertion/elimination is a common path for isomerization or H-shifting reactions



3. Reactivity of olefin complexes

b. C-H activation



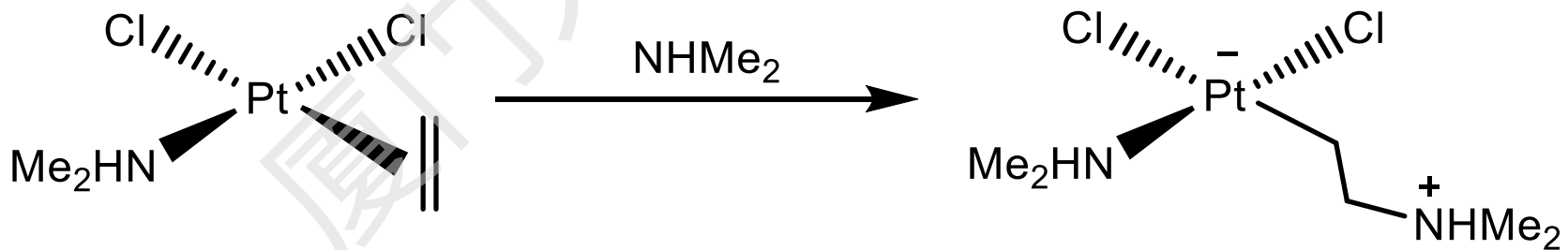
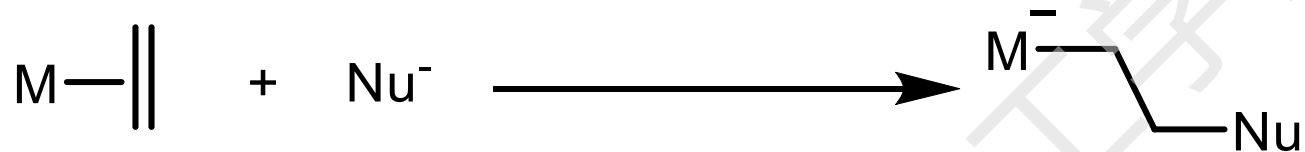
16 e⁻, Mo(0), d⁸

18 e⁻, Mo(II), d⁶



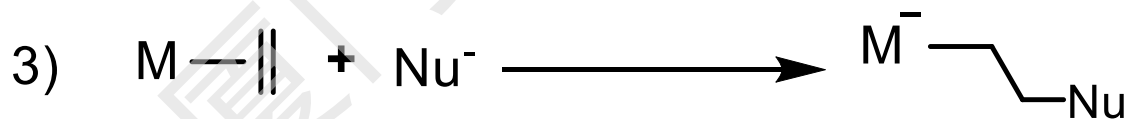
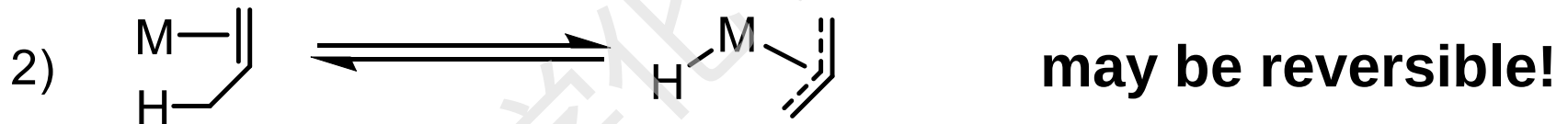
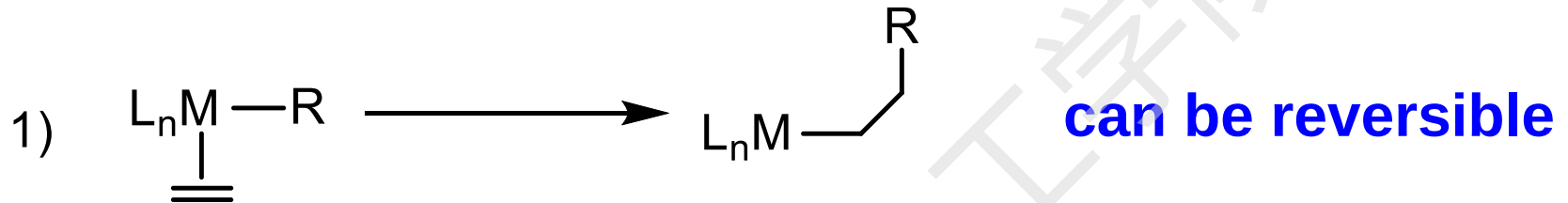
3. Reactivity of olefin complexes

c. Nucleophilic attack on coordinated olefins.





Summary: Reactions of olefin complexes





Practice Problem

J | A | C | S
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Perspective

Palladium(0) π -Lewis Base Catalysis: Concept and Development

Zhi-Chao Chen, Qin Ouyang, Wei Du,* and Ying-Chun Chen*



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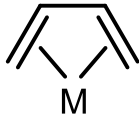
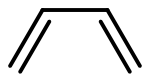
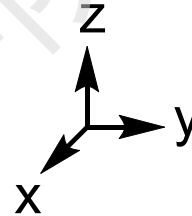
ABSTRACT: The development of a new catalytic strategy plays a vital role in modern organic chemistry since it permits bond formation in an unprecedented and more efficient manner. Although the application of preformed metal complexes as π -base-activated reagents have enabled diverse transformations elegantly, the concept and strategy by directly utilizing transition metals as efficient π -Lewis base catalysts remain underdeveloped, especially in the field of asymmetric catalysis. Here, we outline our perspective on the discovery of palladium(0) as an efficient π -Lewis base catalyst, which is capable of increasing the highest occupied molecular orbital (HOMO) energy of both electron-neutral and electron-deficient 1,3-dienes and 1,3-enynes upon flexible η^2 -complexes formed in situ and resultant π -backdonation. Thus, fruitful carbon–carbon-forming reactions with diverse electrophiles can be achieved enantioselectively in a vinylogous addition pattern, which is conceptually different from the classical oxidative cyclization mechanism. Emphasis will be given to the concept and mechanism elucidation, catalytic features, and reaction design together with perspective on the further development of this emerging field.



B. Butadiene complexes

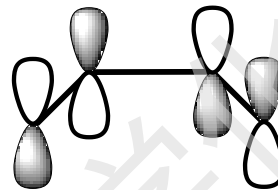
1. common structures of butadiene complexes

MO of π system of a butadiene ligand

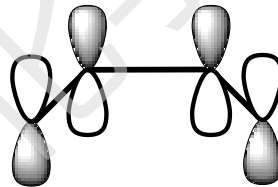


most common

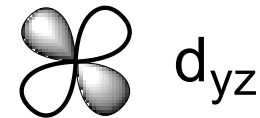
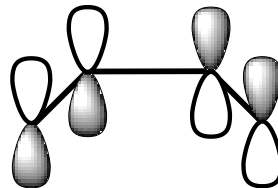
4π —



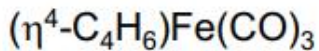
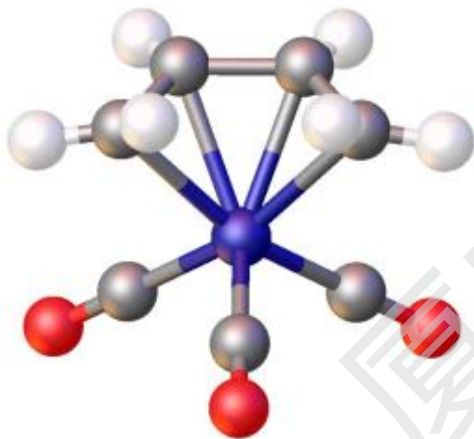
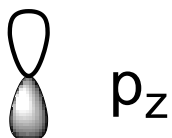
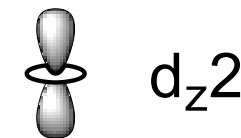
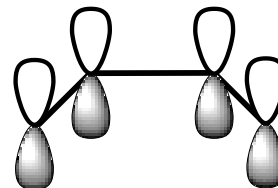
3π —



2π $\uparrow\downarrow$



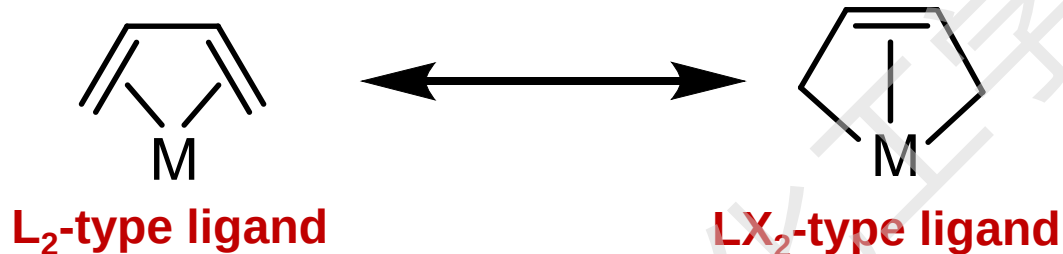
1π $\uparrow\downarrow$



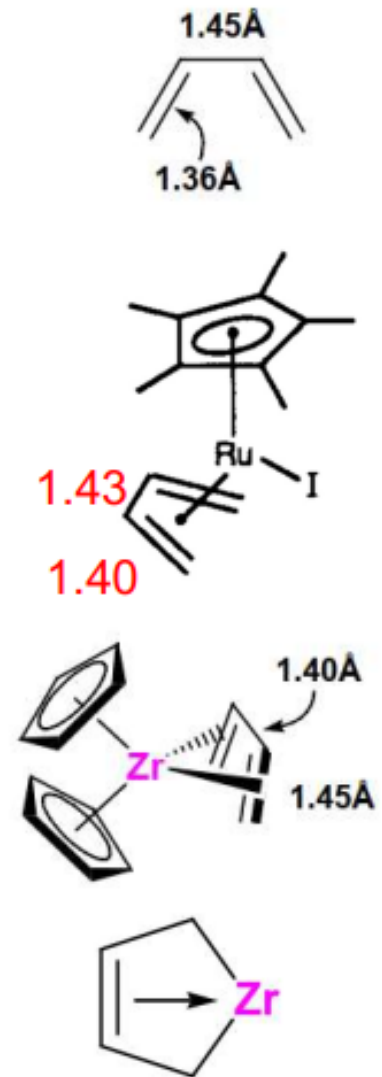
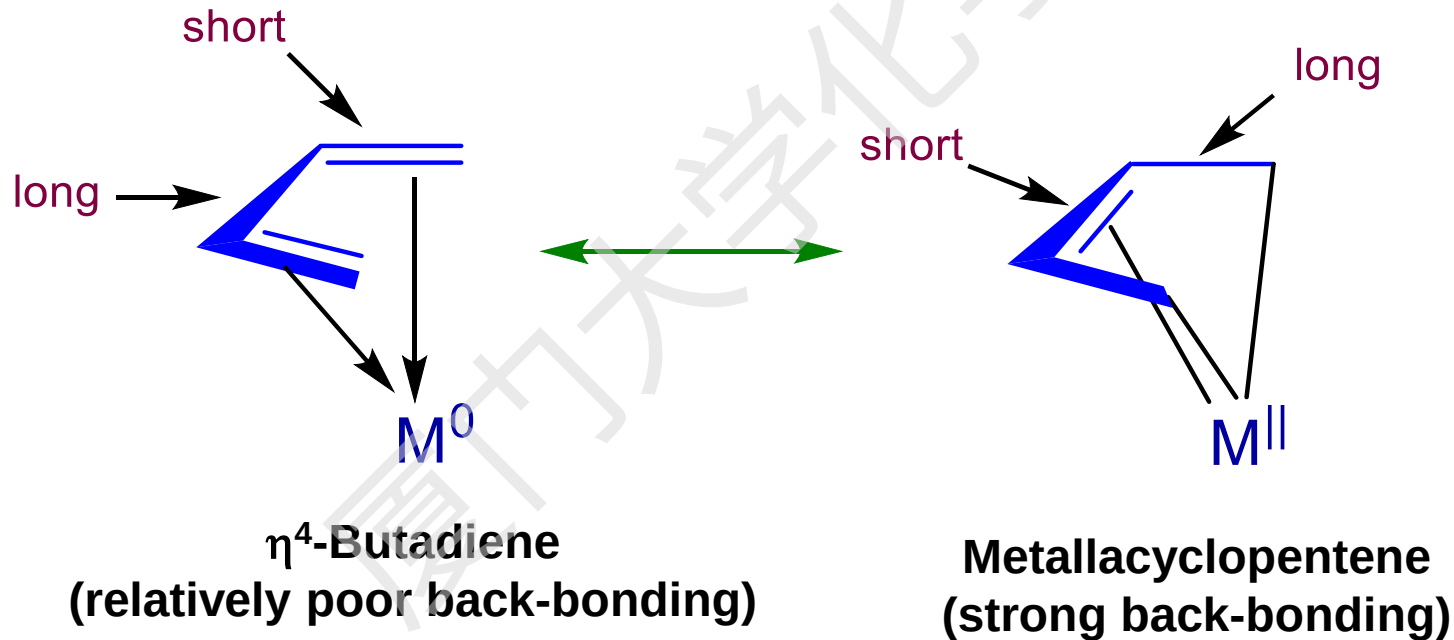


B. Butadiene complexes

Resonance structures of butadiene complexes



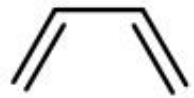
Resonance forms





B. Butadiene complexes

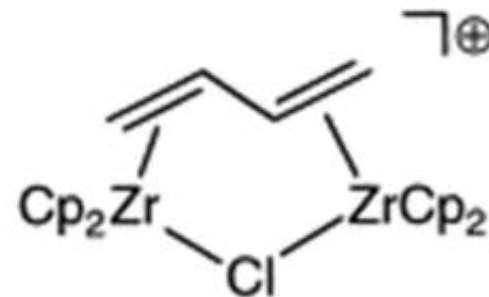
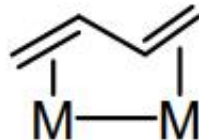
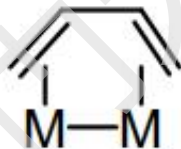
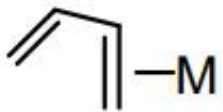
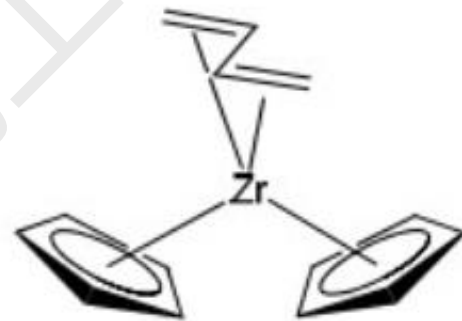
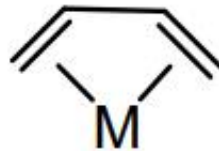
Other coordination modes of butadienes (**for information**)



s-cis



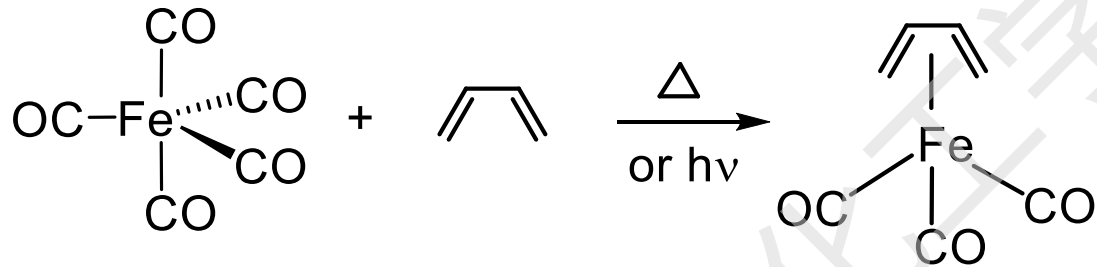
s-trans



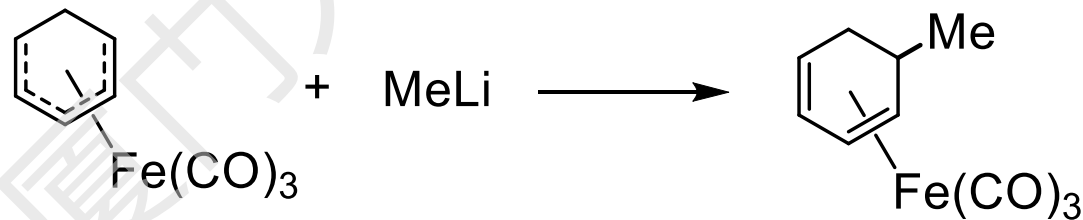
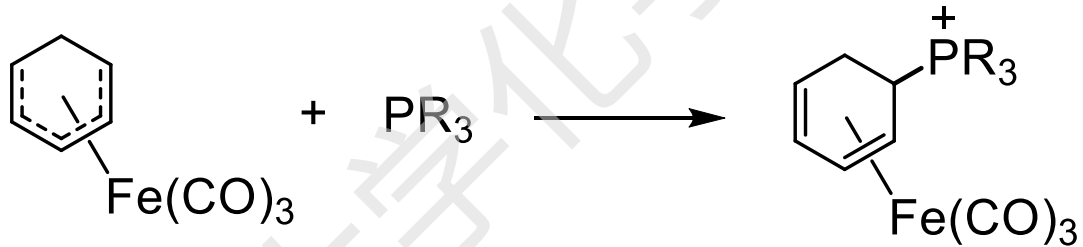


2. Synthesis of butadiene complexes

a. By substitution reactions.



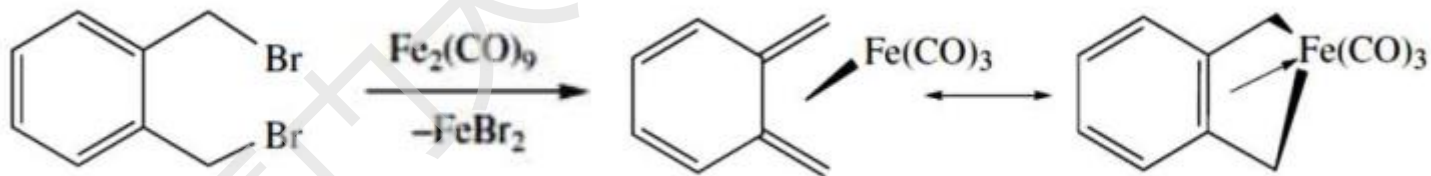
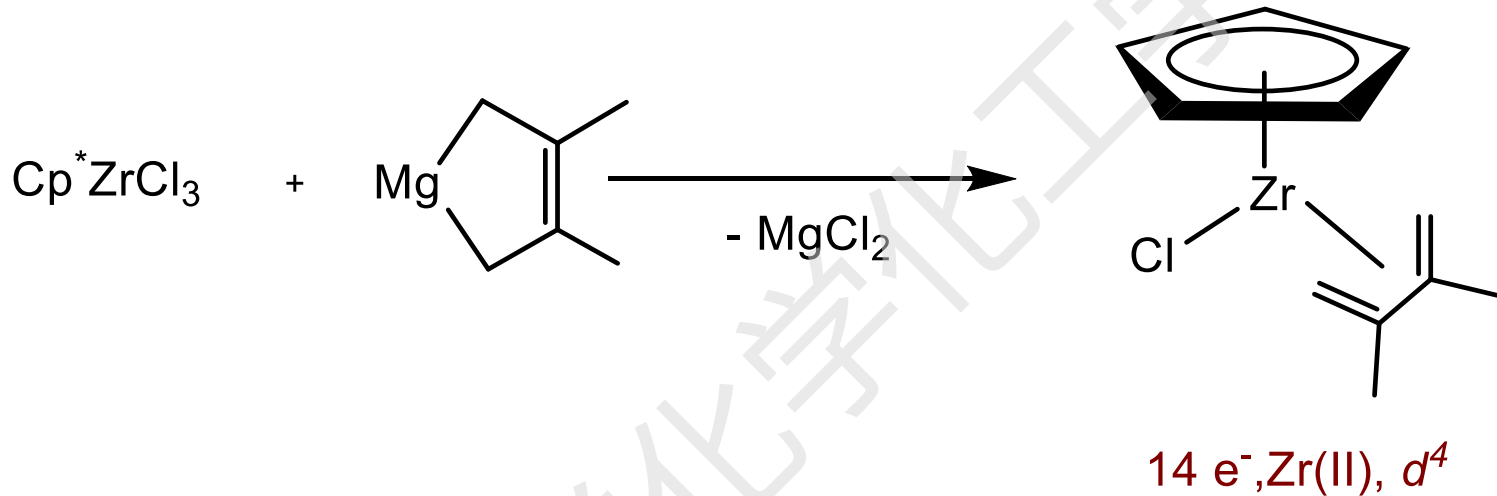
b. Reaction of coordinated ligands.





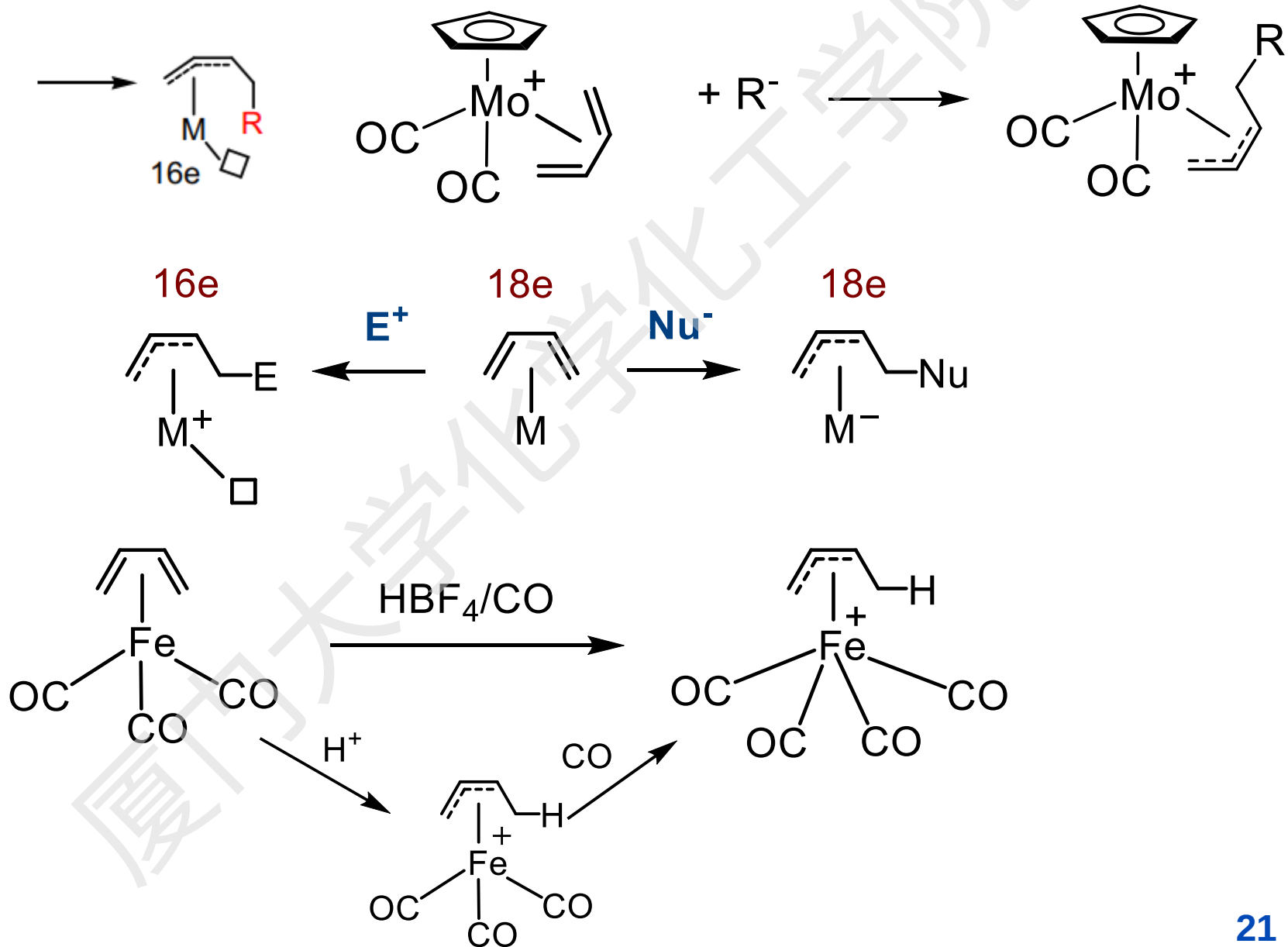
2. Synthesis of butadiene complexes

c. Other methods





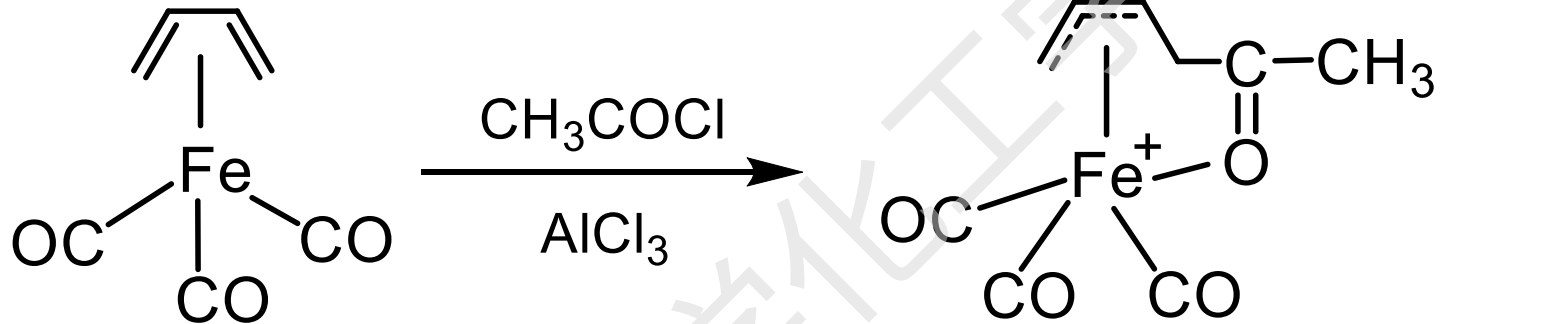
3. Reactions of butadiene complexes





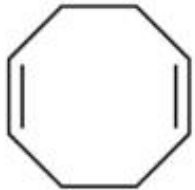
Practice Problems

Mechanism?

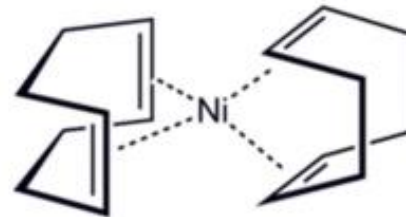
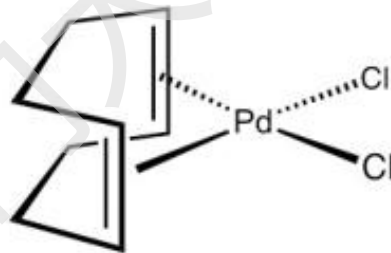
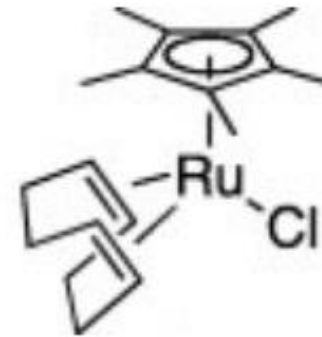
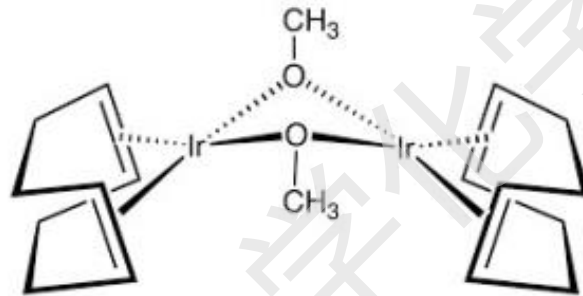
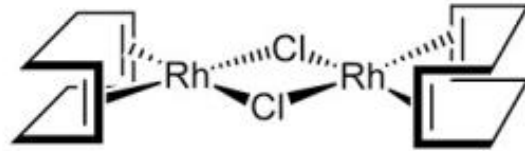




Related diene complexes



1,5-cyclooctadiene
(COD)



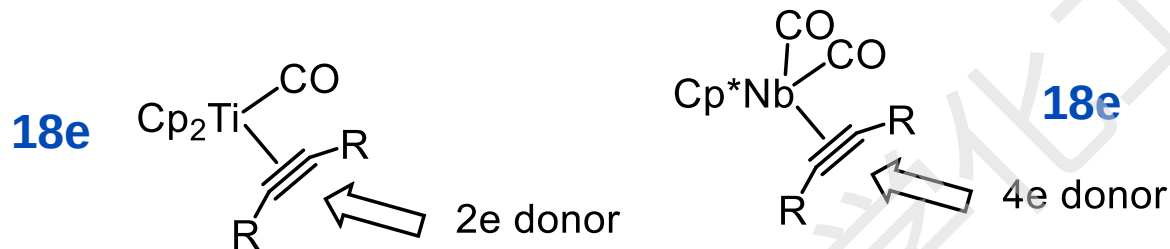


C. Alkyne complexes

1. Structure and bonding of alkyne complexes

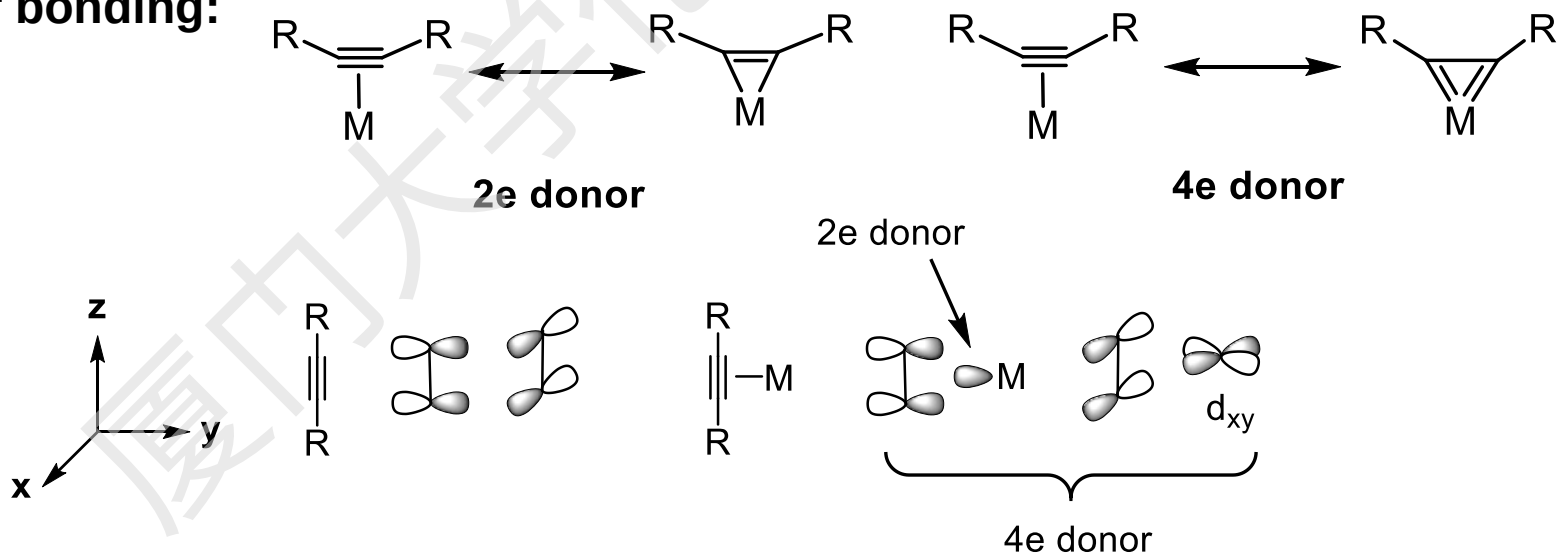
$R-C\equiv C-R$ can be simple η^2 -ligand or bridging ligand.

As simple η^2 -ligand, they can be 2e or 4e donors. e.g.



How can alkynes be 2e or 4e donors?

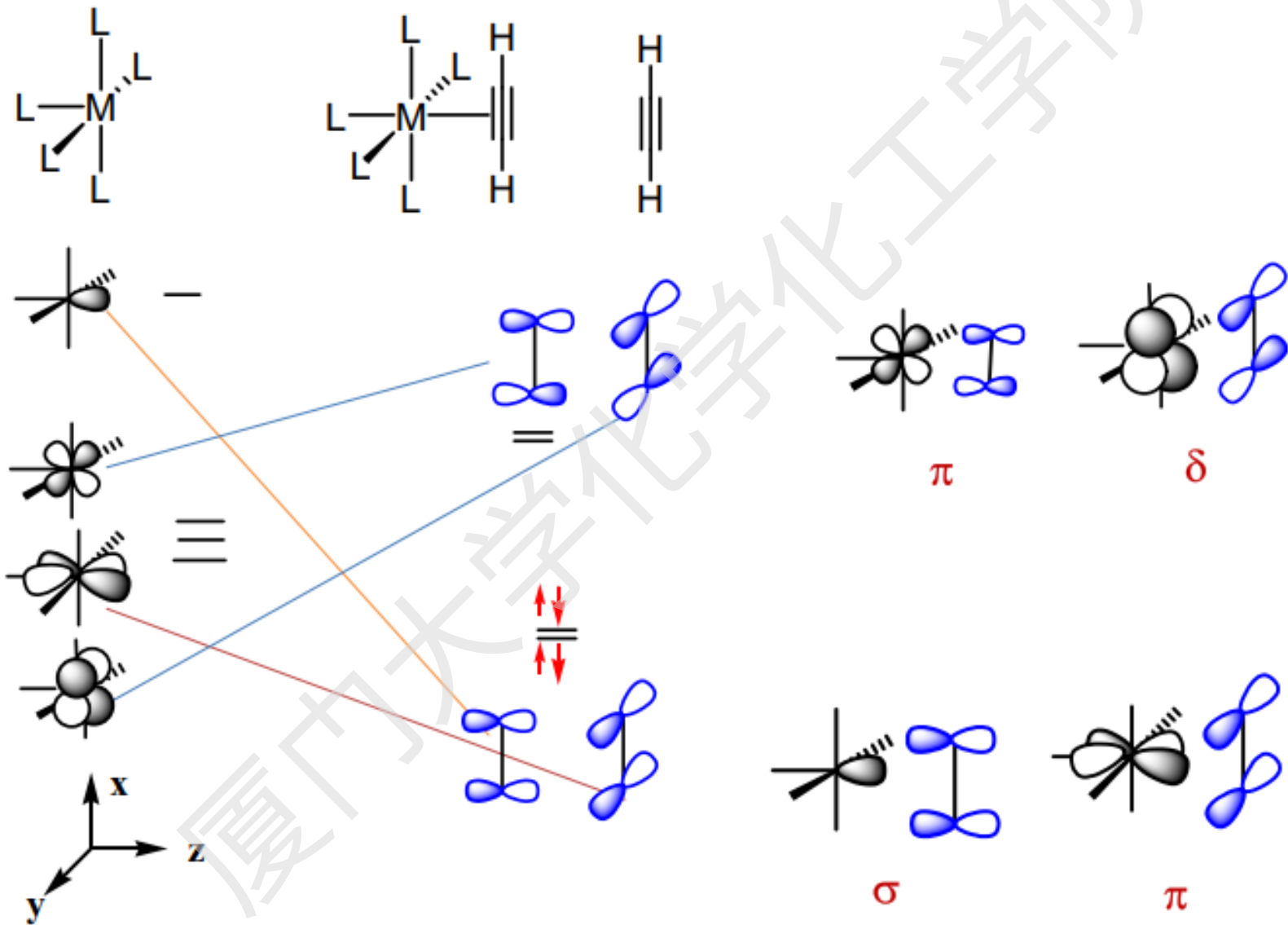
In terms of bonding:



Alkynes are more electronegative, therefore are better π -acceptors

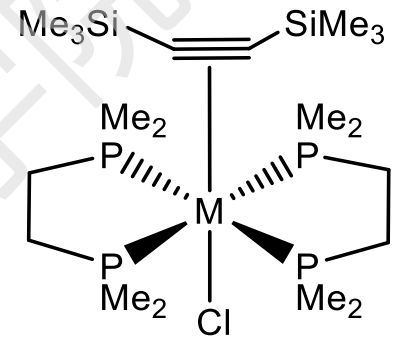
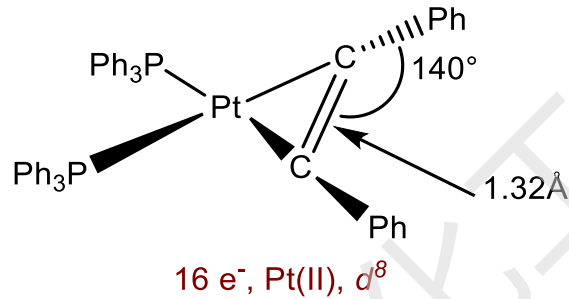
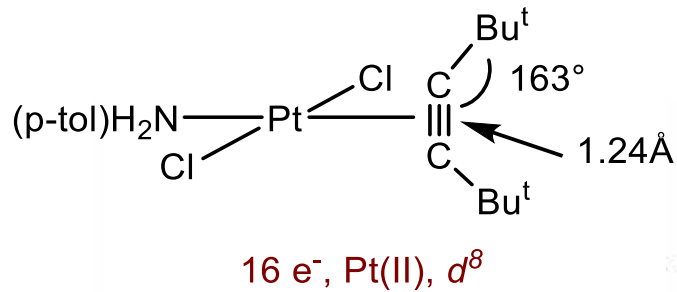


Possible bonding interactions between ML_5 fragment and an alkyne





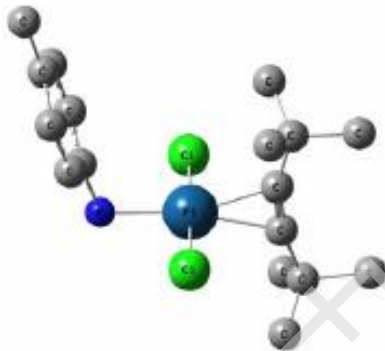
Examples of structures of Alkyne complexes



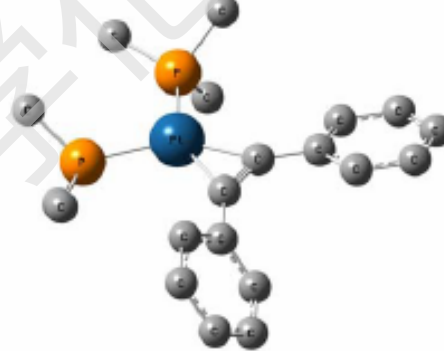
perpendicular coordination to Pt^{II};
C≡C short, wide C-C-R angle

in-plane bonding to Pt⁰;
C≡C elongated, smaller C-C-R angle

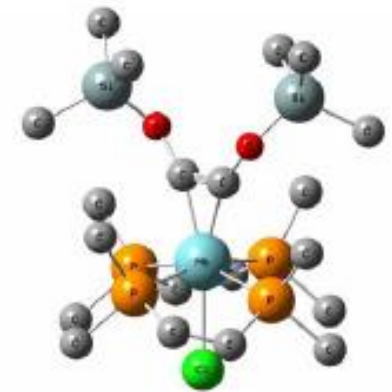
M = Nb, C-C 1.31 Å
M = Ta, C-C 1.34 Å



2e donor alkyne



$\text{H}-\text{C}\equiv\text{C}-\text{H}$
1.203 Å

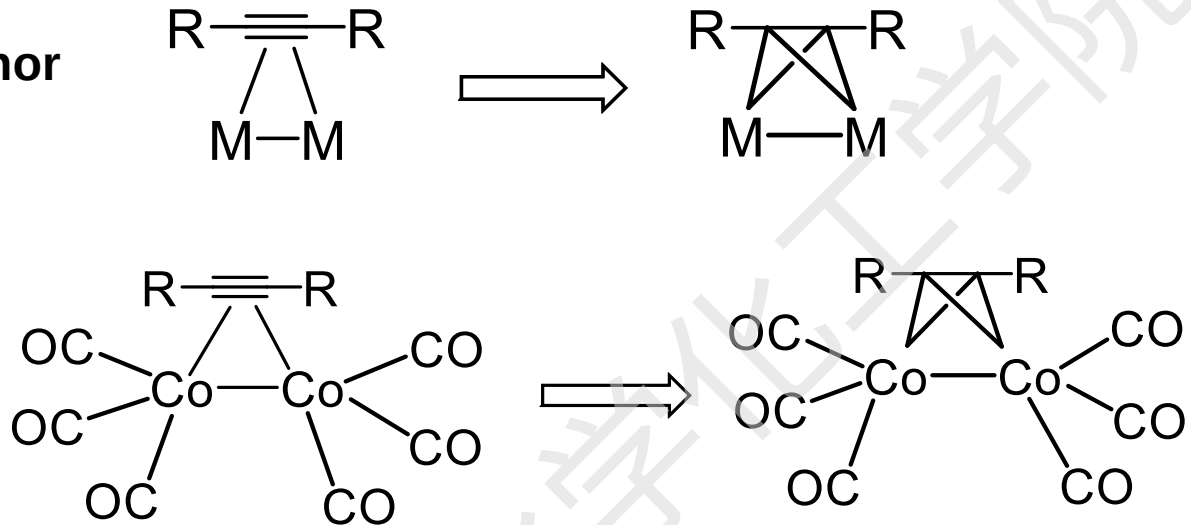


4e donor alkyne

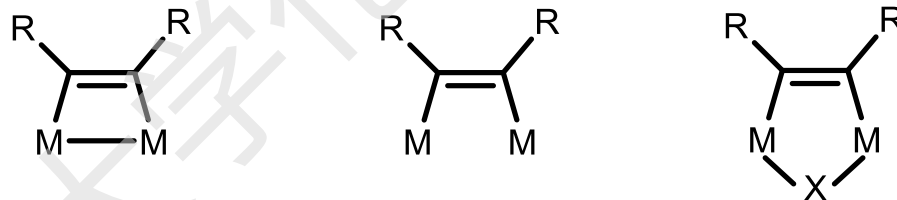


Alkynes as bridging ligands (for information)

4e donor



2e donor



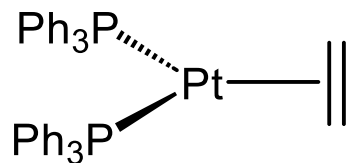
More complicated bonding modes



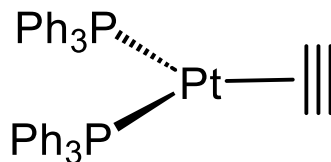


Practice Problems

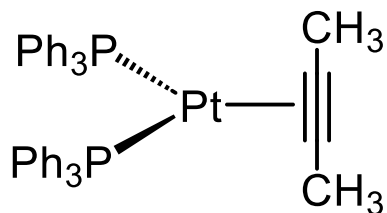
For each pair of the following complexes, which one is more stable?



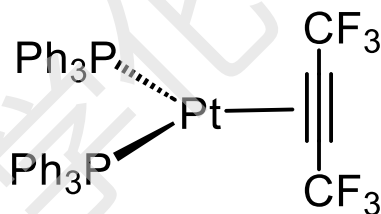
(a)



(b)



(a)

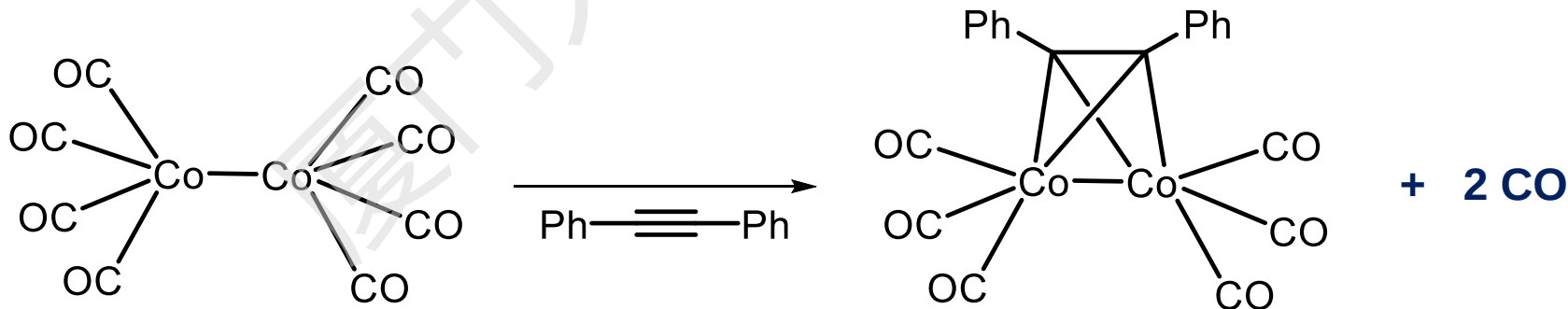
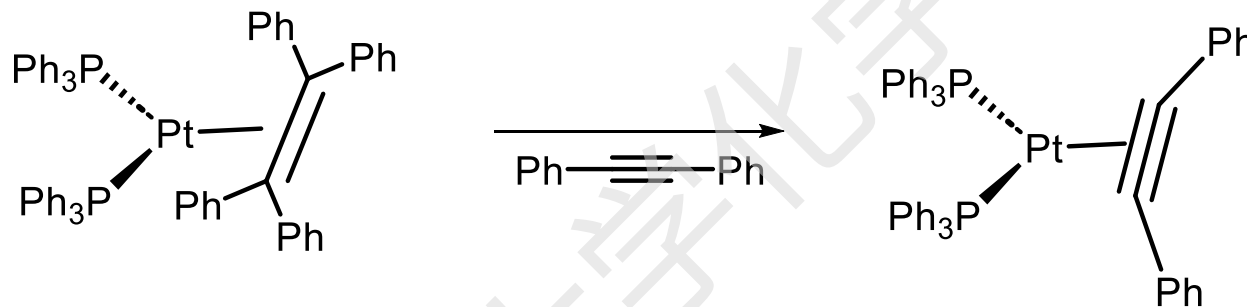
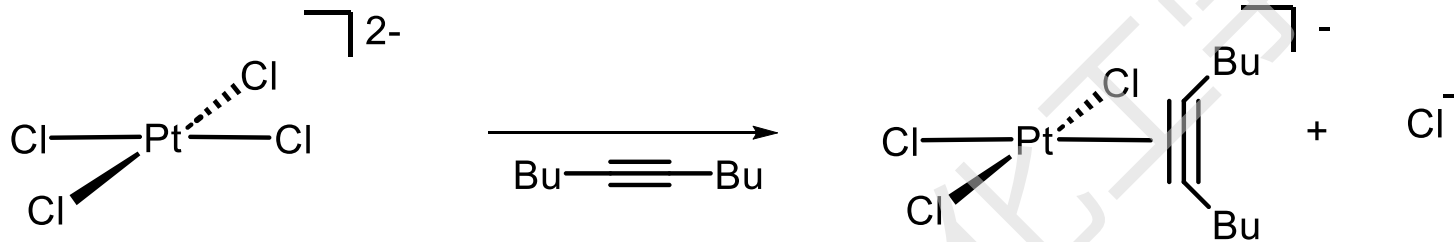


(b)



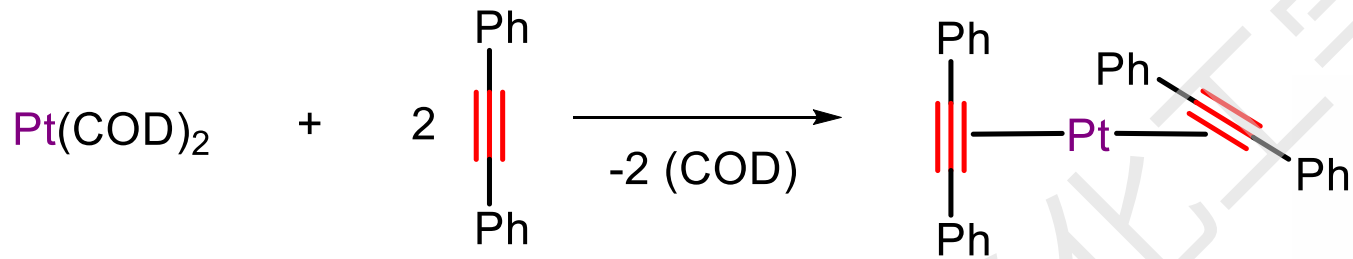
2. Preparation of alkyne complexes: similar to olefin complexes

a. *By substitution reactions: e.g.*





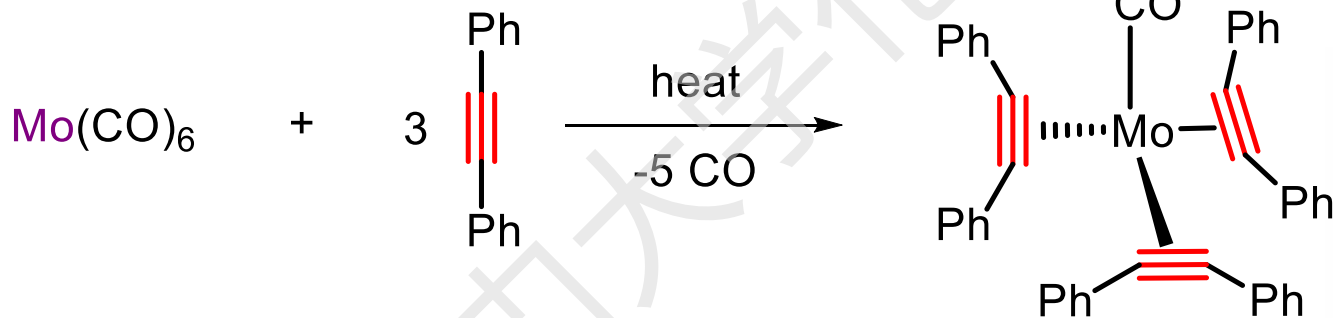
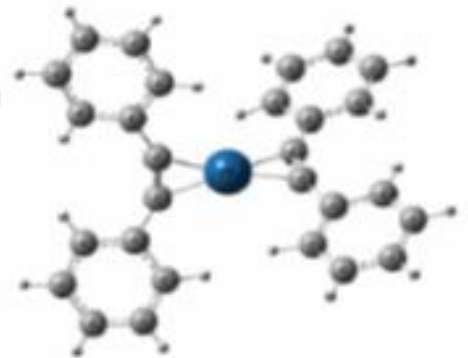
Additional examples



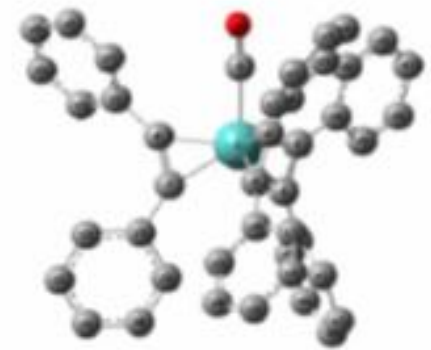
Two 4e donor alkynes

C-C 1.29 Å
Pt-C 2.02 Å

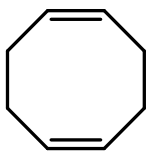
18 e⁻, Pt(0), d¹⁰



18 e⁻, Mo(0), d⁶



Two 4e donor alkynes
one 2e donor alkyne

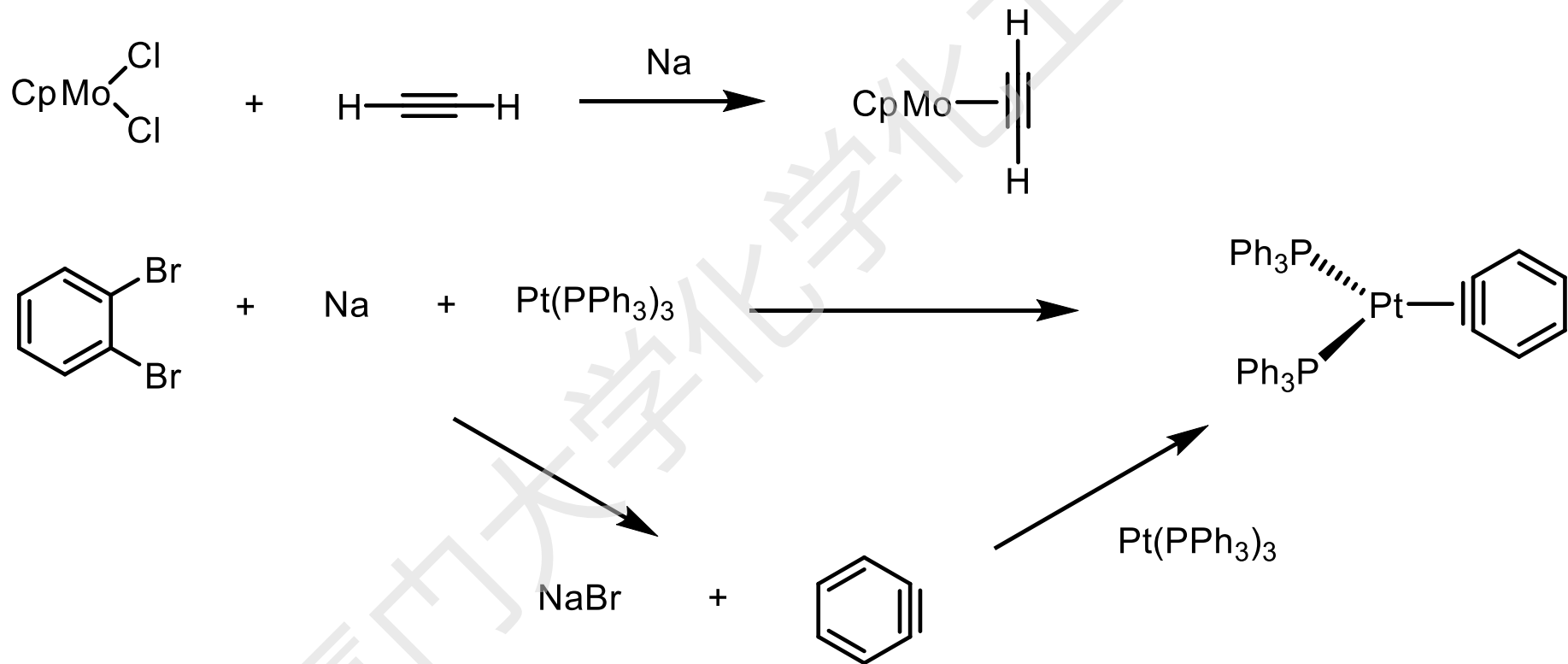


COD



2. Preparation of alkyne complexes: similar to olefin complexes

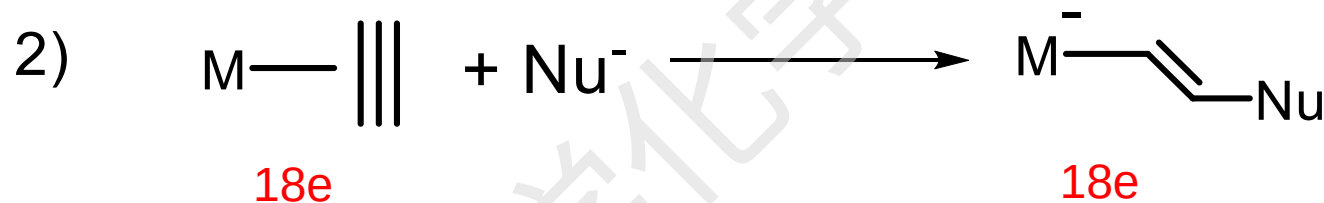
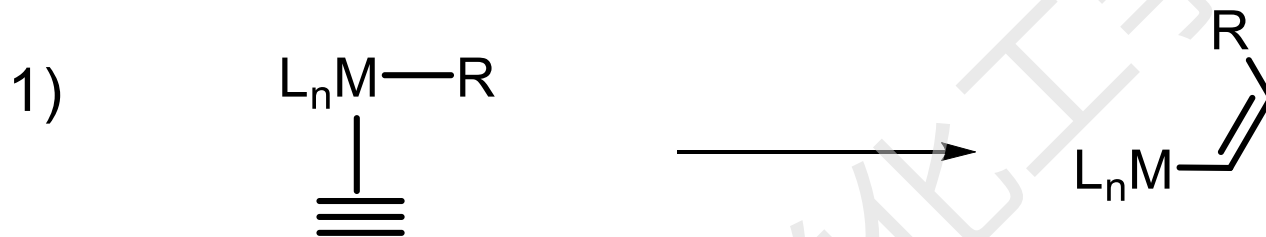
b. By other methods: e.g.



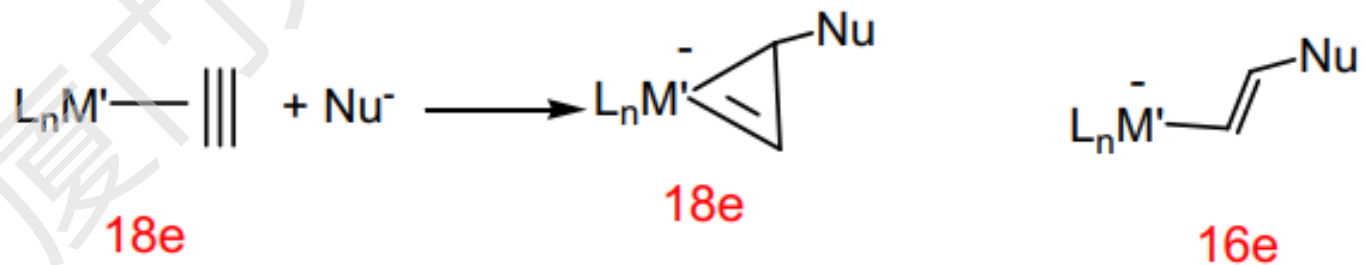


3. Reactivity of alkyne complexes

Similar to olefins complexes



2e donor alkynes

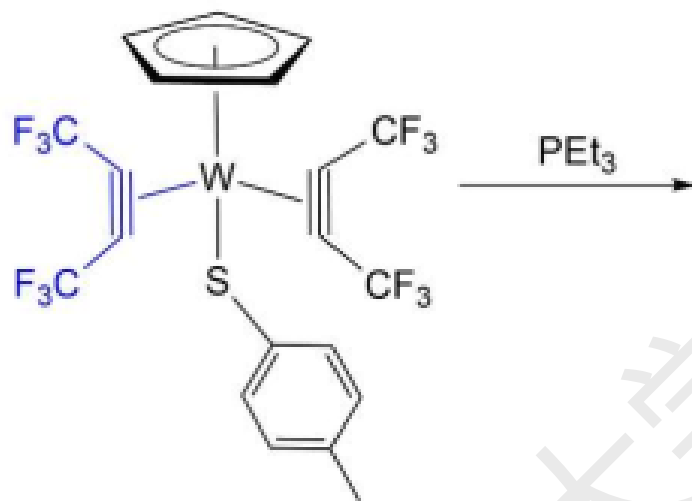
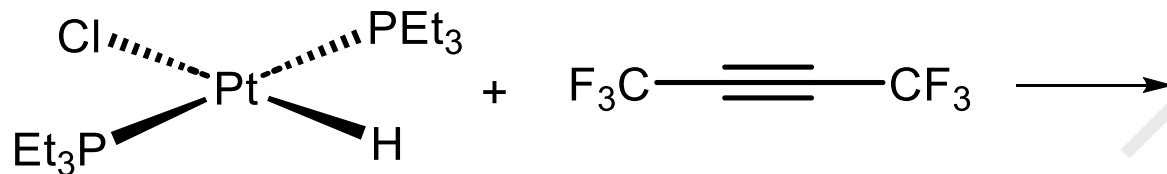


4e donor alkynes



Practice Problems

Give the product for the following reaction.

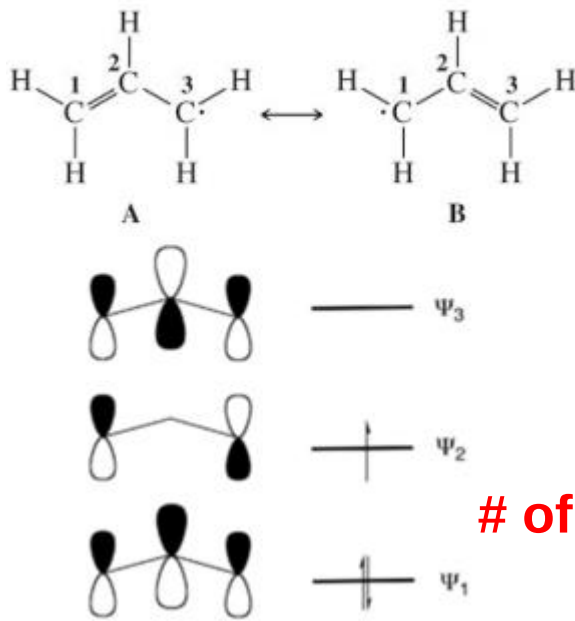




D. Enyl Complexes

Enyl: Organic ligands with odd number of carbon chains.

π MOs of allyl radical

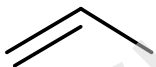


of π e $^-$: 3e



C_3H_5

or



M

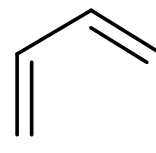
η^3 -allyl

η^3 - C_3H_5

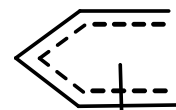


C_5H_7

or

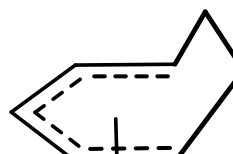


5e



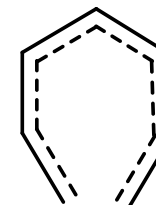
M

η^5 -pentadienyl



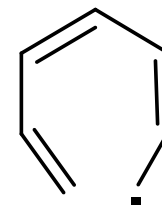
M

η^5 - C_7H_9

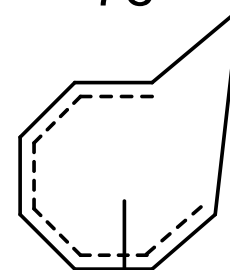


C_7H_9

or



7e



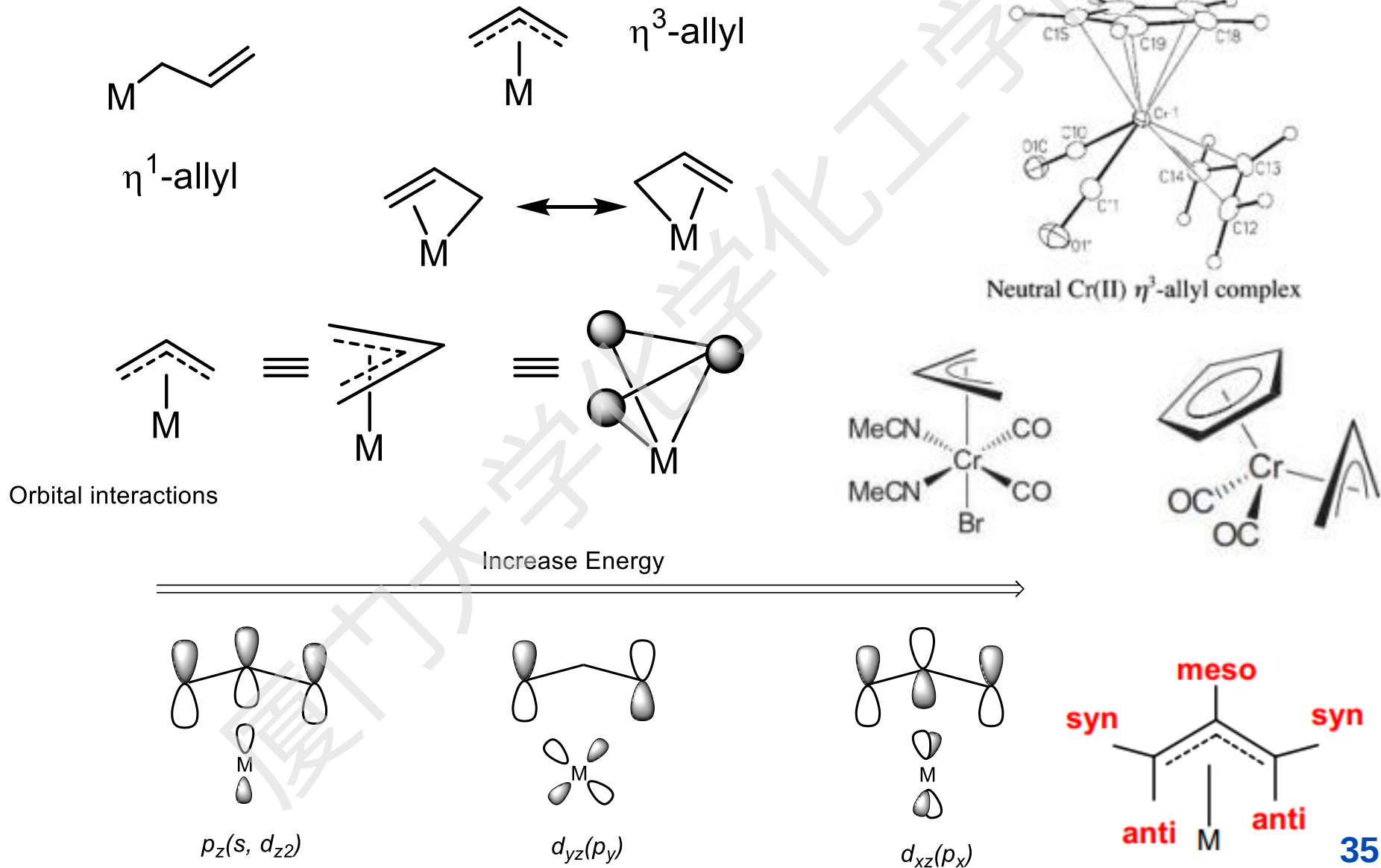
M

η^7 - C_8H_9 **34**



1. Allyl complexes

a. Structure and bonding of allyl complexes

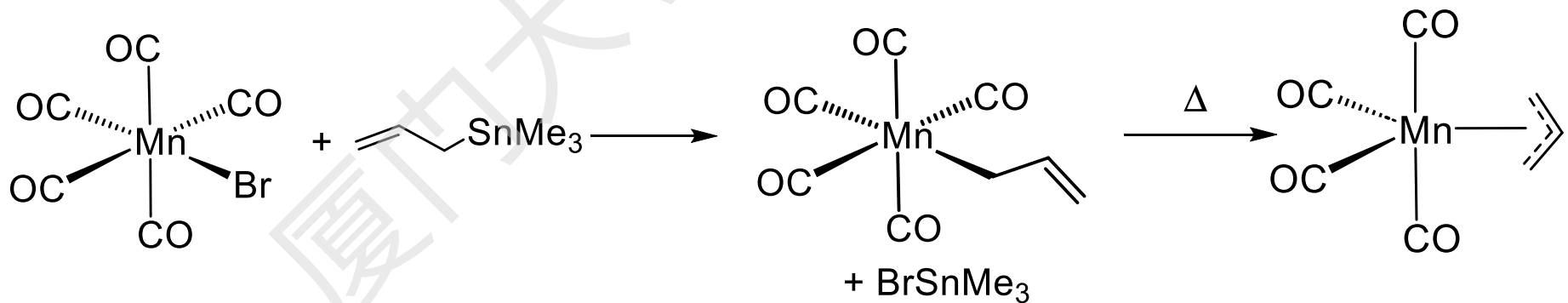
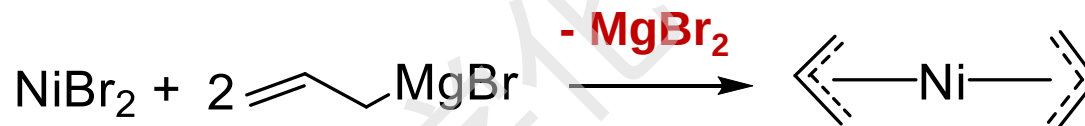




1. Allyl complexes

b. Preparation of allyl complexes

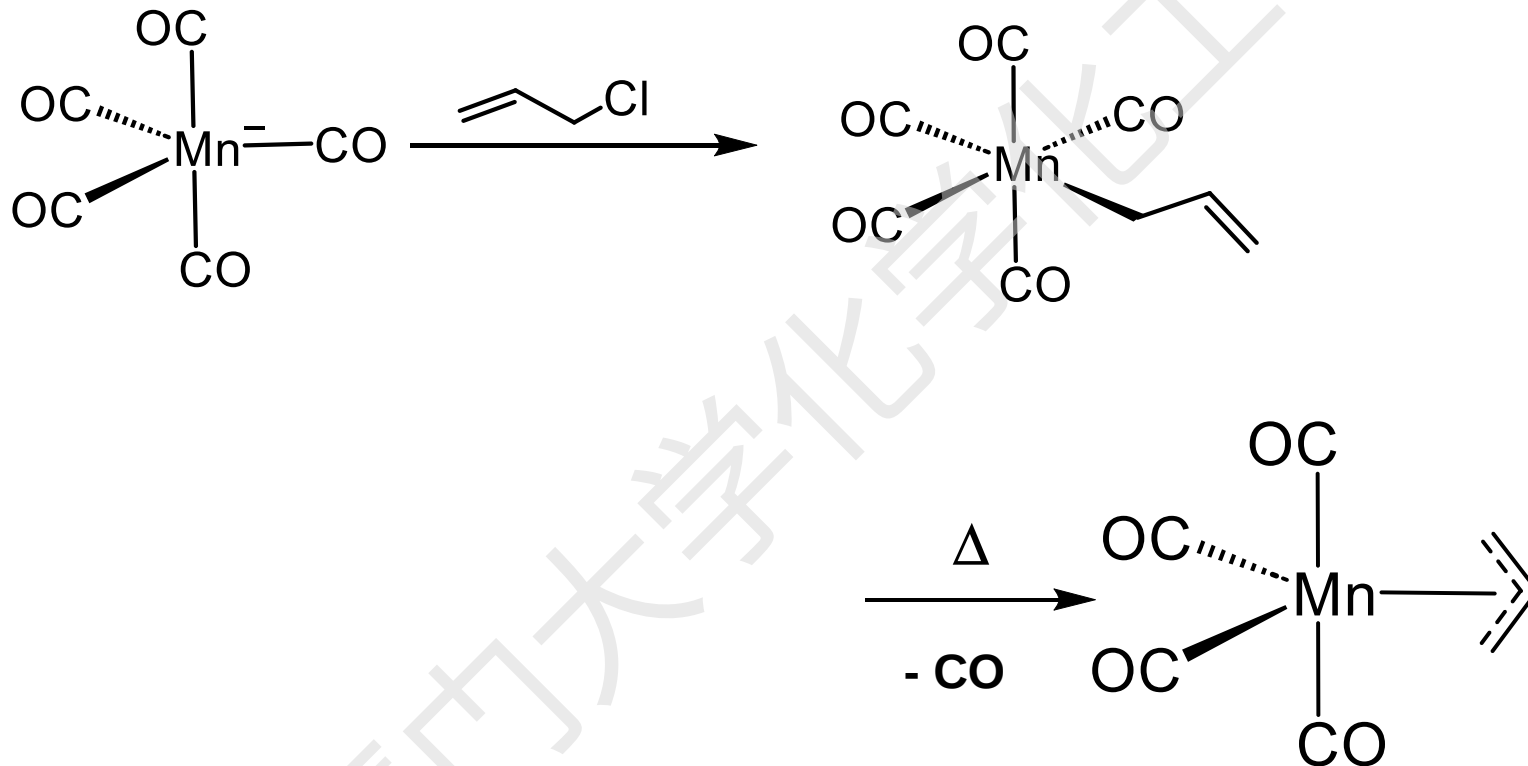
i) Metal salt + main group organometallic reagents





b. Preparation of allyl complexes

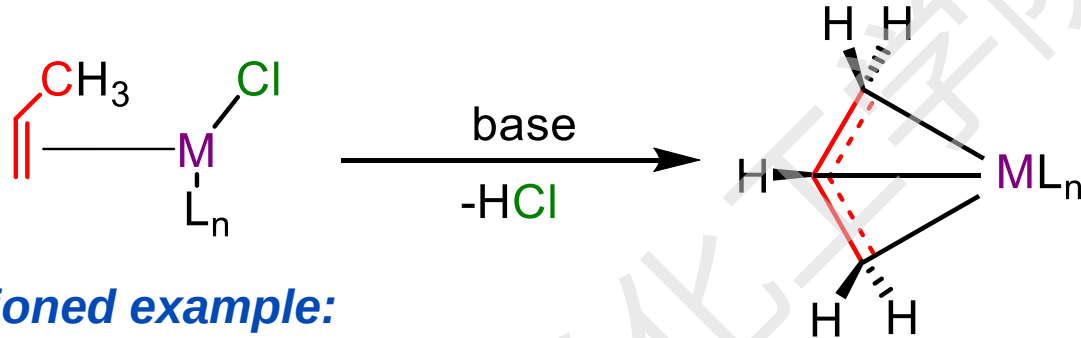
ii) Low valent metal complexes + allylhalide



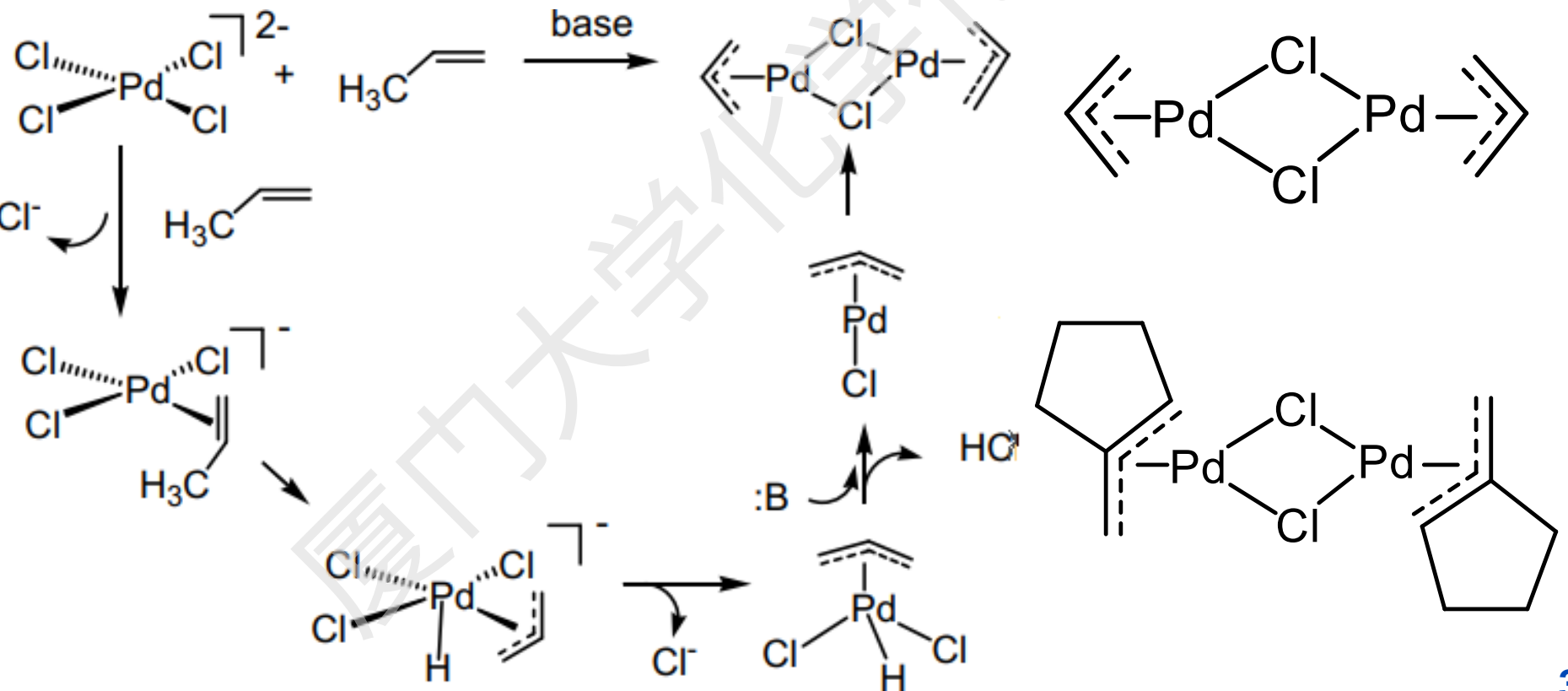


b. Preparation of allyl complexes

iii) Metal salts + olefin + base.



A previously mentioned example:



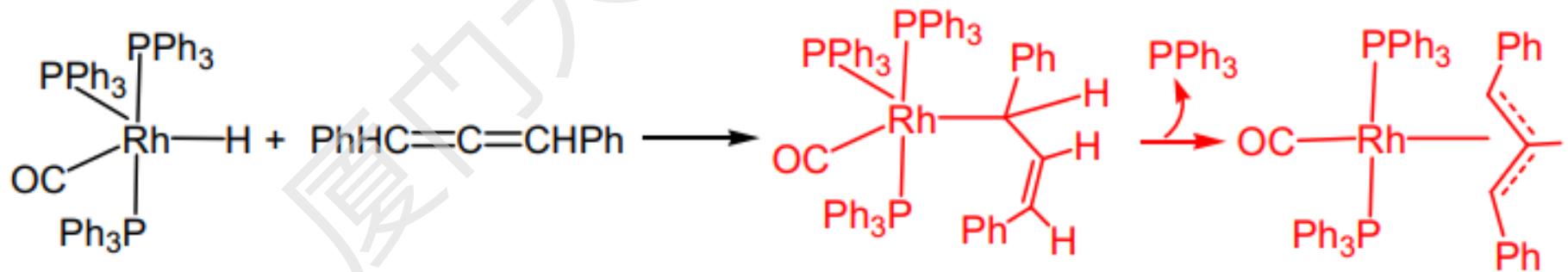
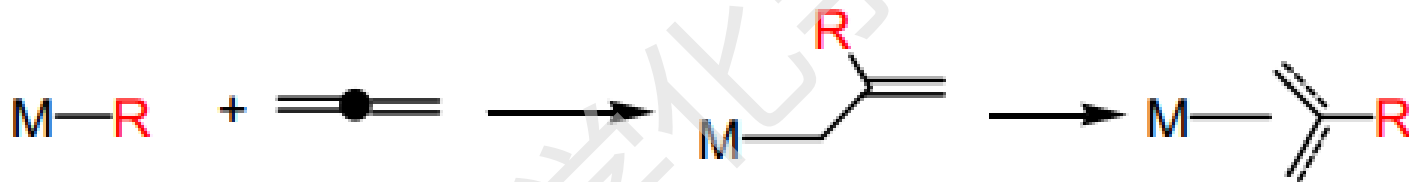
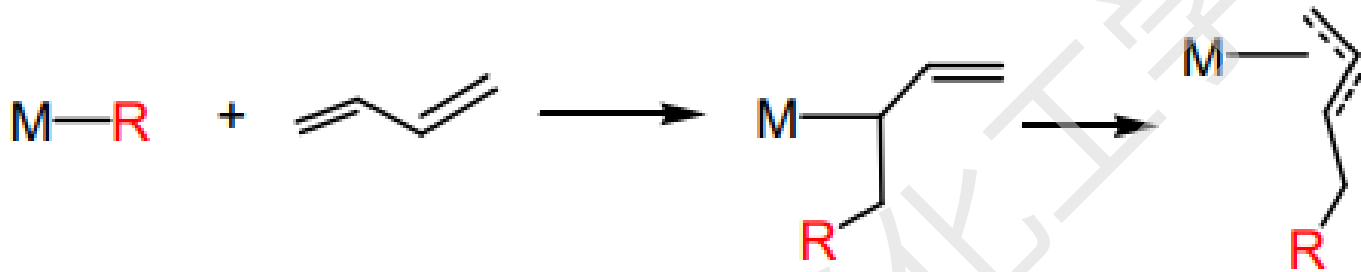


The diagram illustrates the catalytic cycle for the hydroformylation of ethylene using a rhodium complex. The cycle starts with a 16-electron rhodium complex (Rh⁺), which adds ethylene to form an 18-electron intermediate. This intermediate then adds CO to form a 19-electron complex, which is in equilibrium with a 16-electron rhodium complex (Rh⁺) and a carbonyl ligand. The cycle is completed by the addition of H₂ to form a 18-electron complex, which then undergoes reductive elimination to yield the aldehyde product and regenerate the 16-electron catalyst.



b. Preparation of allyl complexes

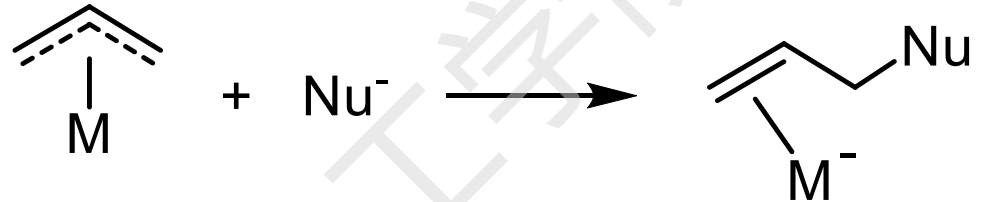
v) Insertion reactions of butadiene or allenes



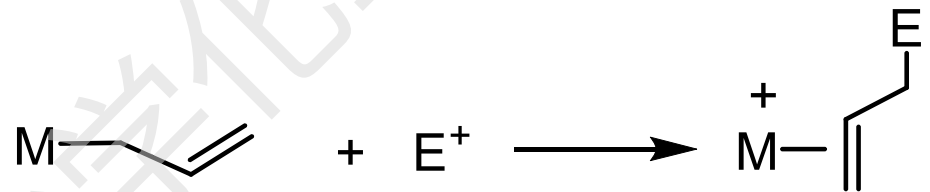


c. Reactivity of allyl complexes

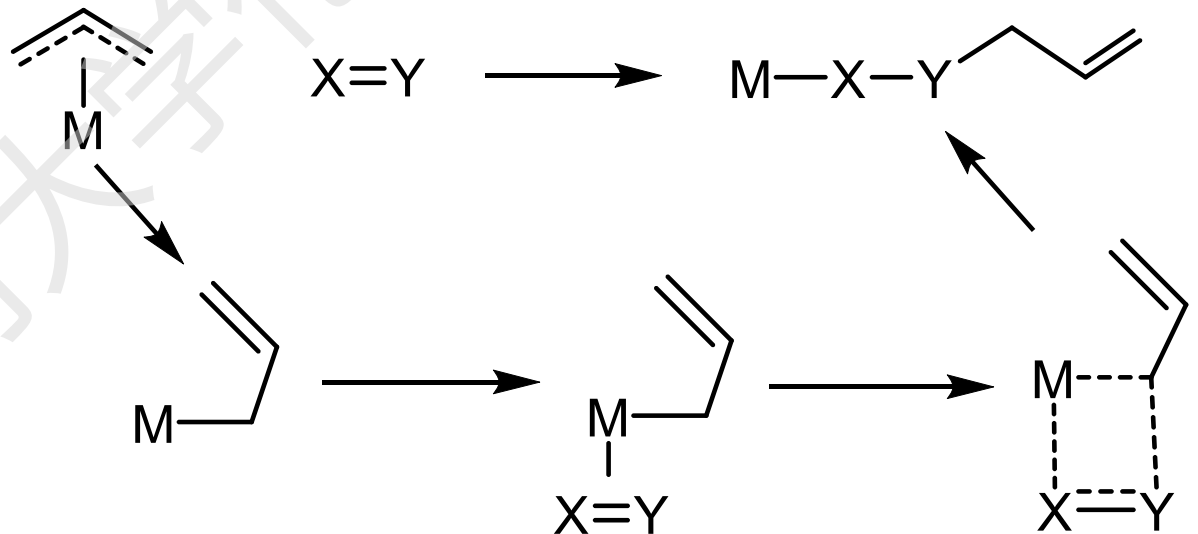
i) With Nucleophiles:



ii) With electrophile (for η^1 -allyls):



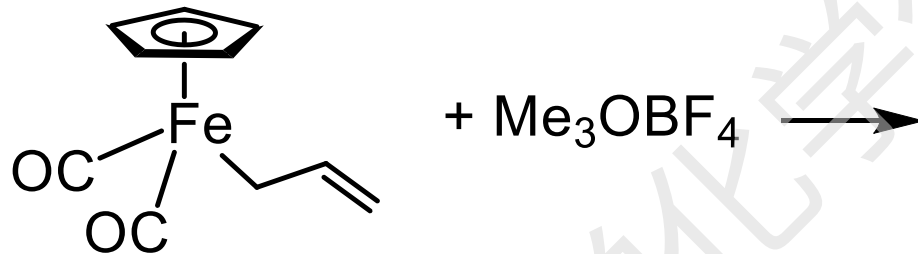
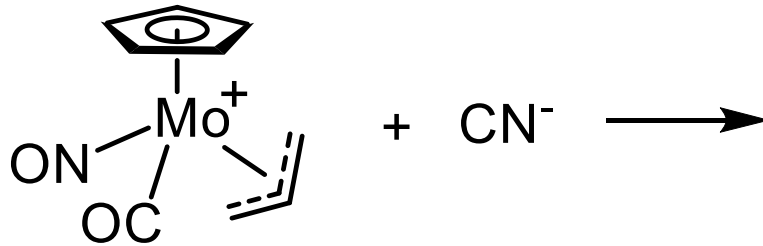
iii) Insertion reactions:





Practice Problems

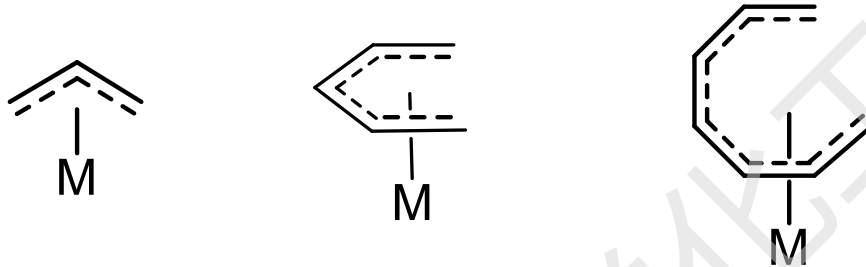
Predict the product of following reactions.



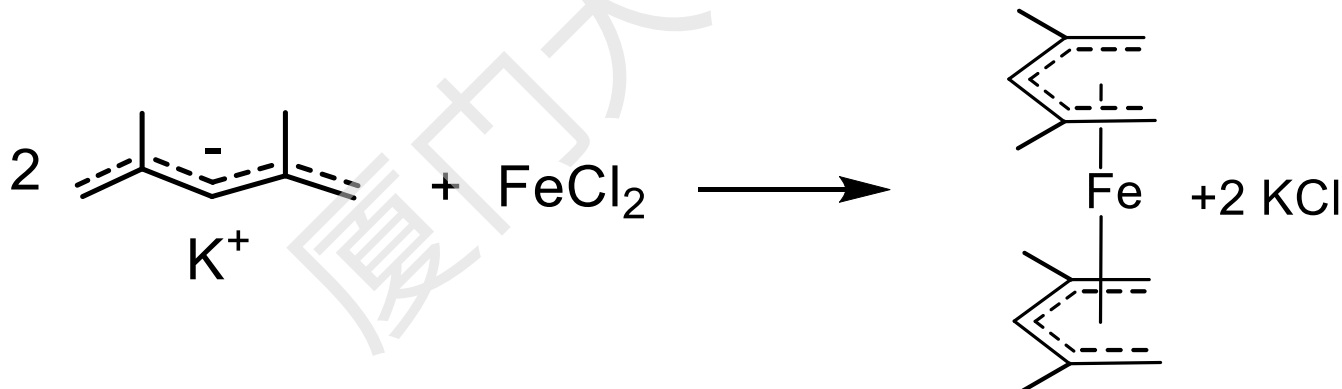
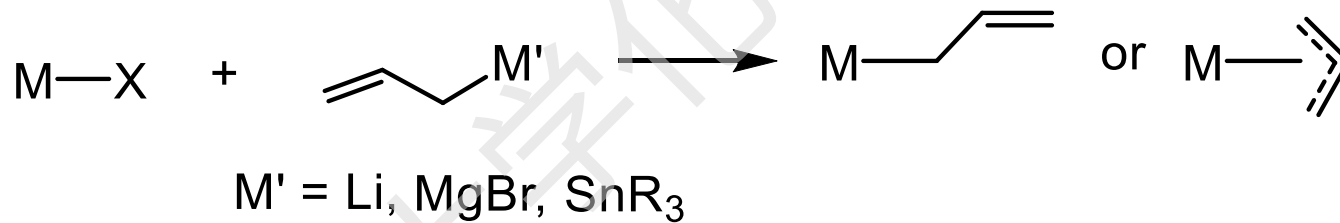


2. Examples of other enyl complexes and their preparation

They can be prepared similarly as allyl complexes.



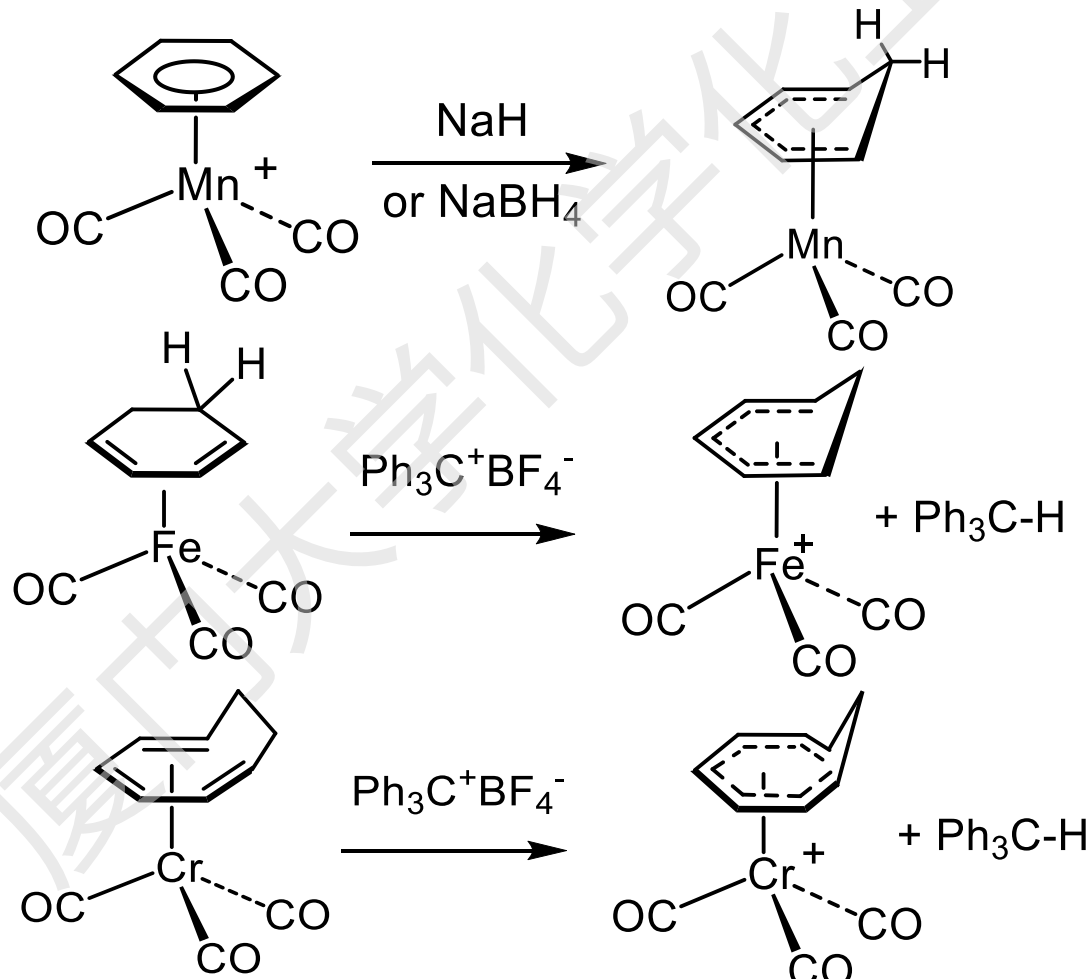
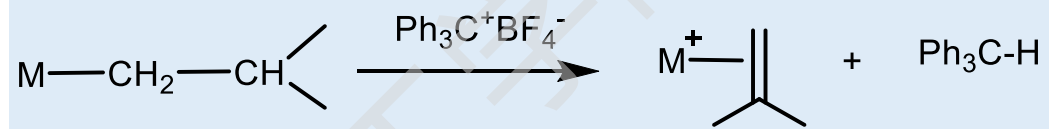
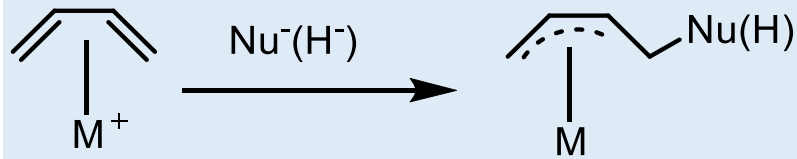
a) Metal salt + main group organometallic reagents.





2. Examples of other enyl complexes and their preparation

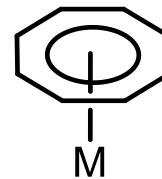
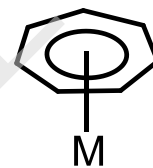
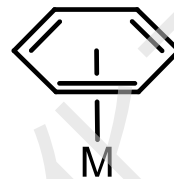
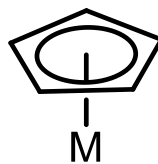
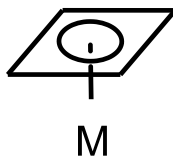
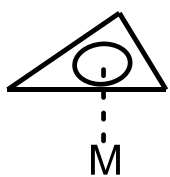
b) From coordinated ligands (via transfer of H^+ or H^-)





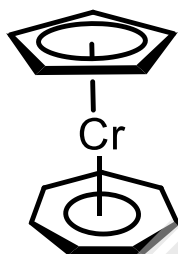
E. Complexes of the cyclic π -perimeters C_nH_n

Common complexes of cyclic π -perimeters C_nH_n :

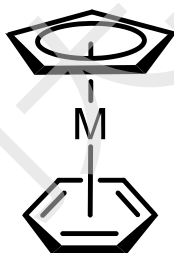


Common structural types

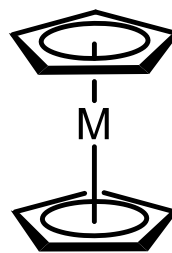
a) Sandwich complexes



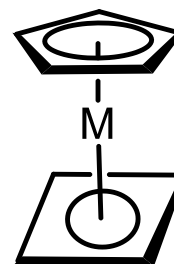
M= Cr



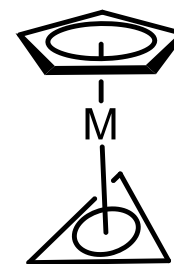
Mn



Fe



Co

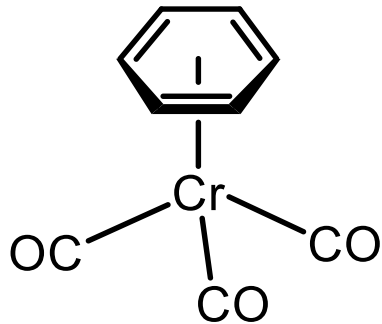


Ni

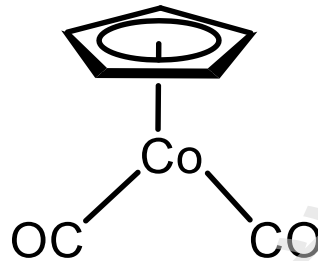


Common structural types

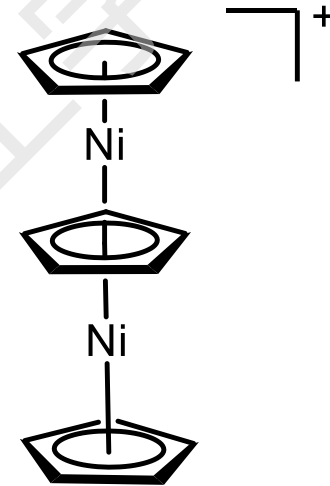
b) Half-sandwich complexes



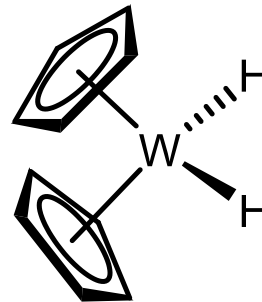
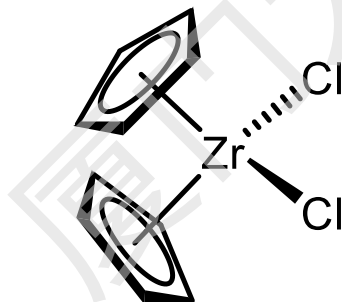
“piano stool”



c) Multidecker sandwich

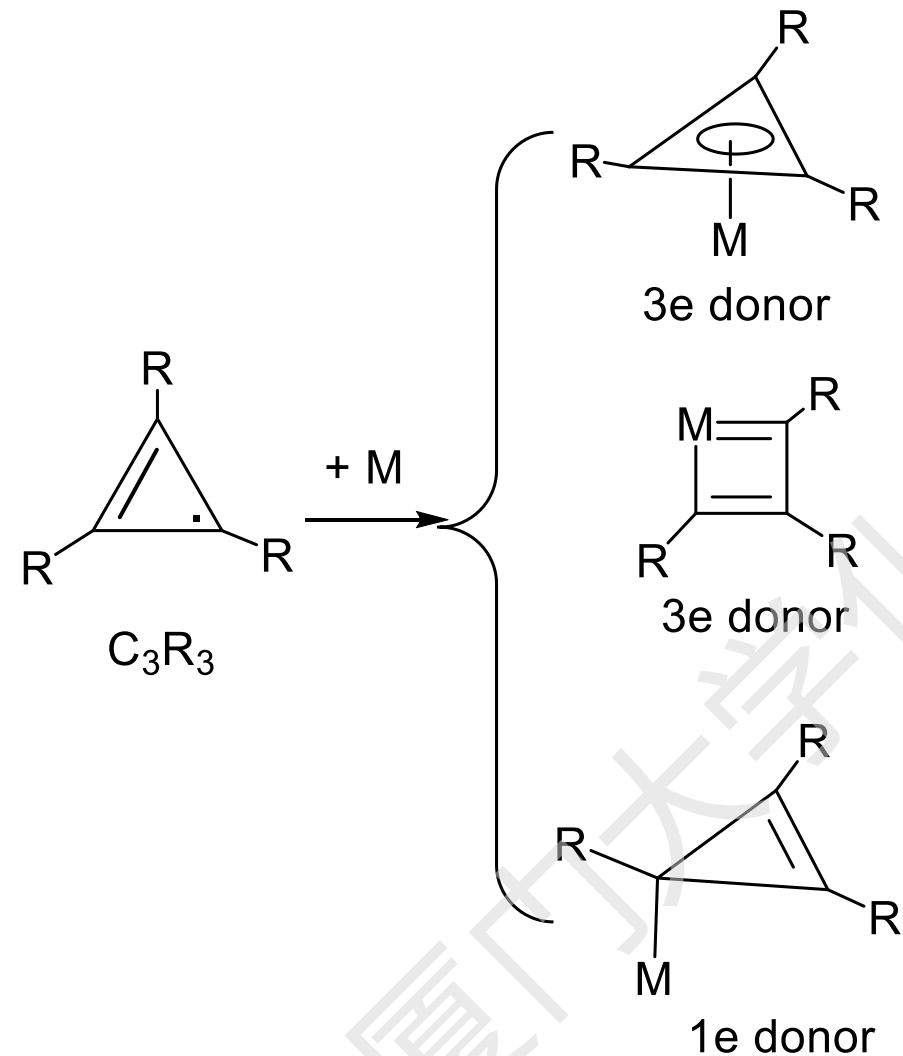


d) Complexes with tilted sandwich structures

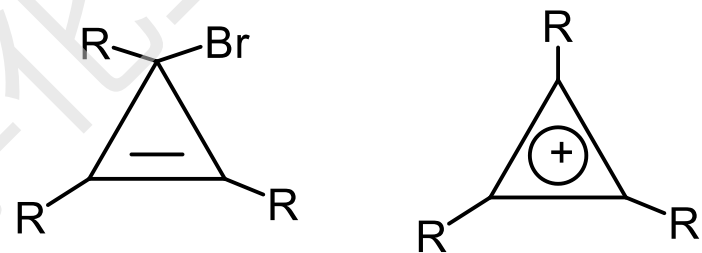




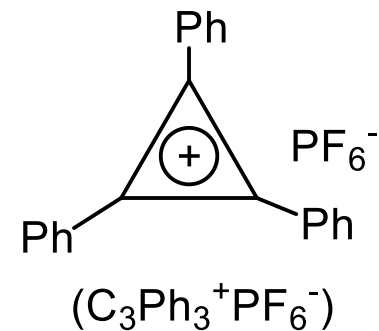
1) C_3R_3 as ligands



Common starting materials for preparing $\eta^3-C_3R_3$ complexes



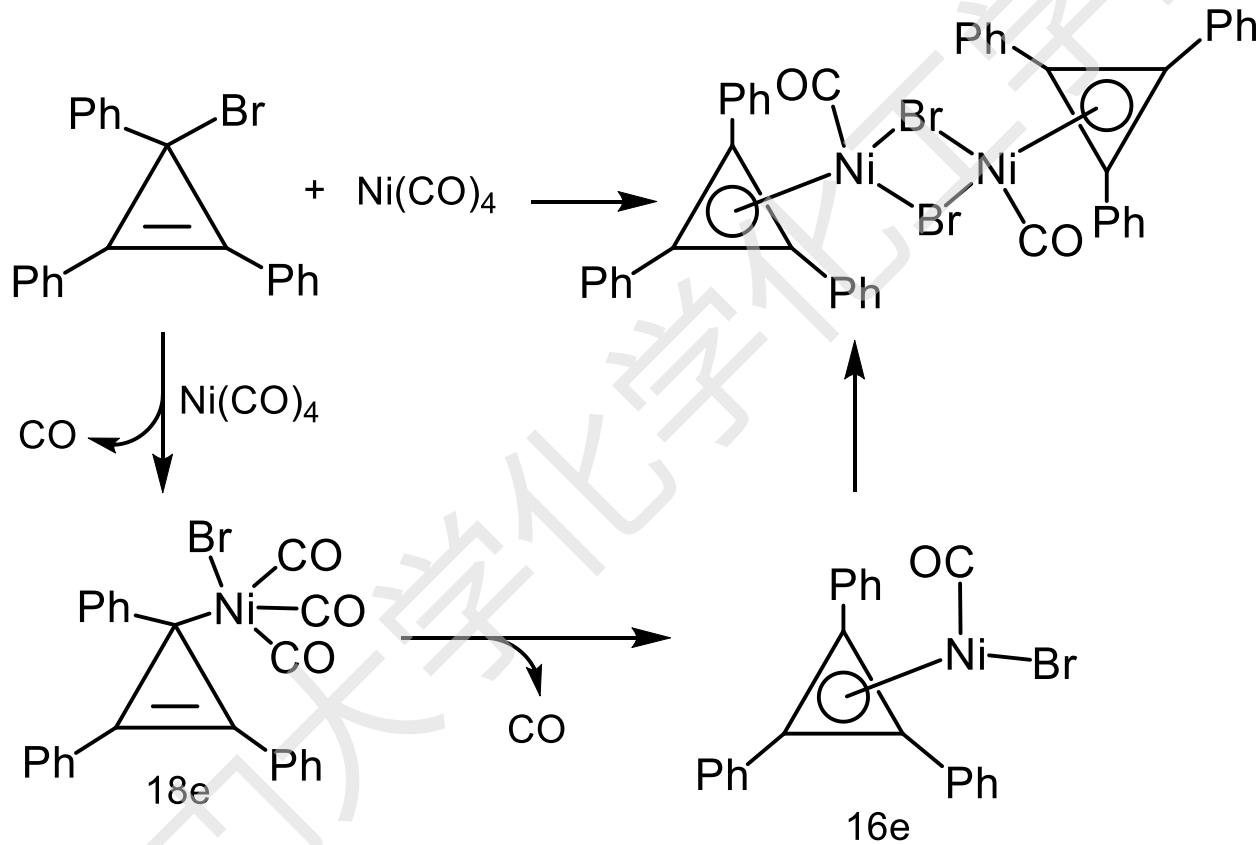
e.g.



2e, aromatic

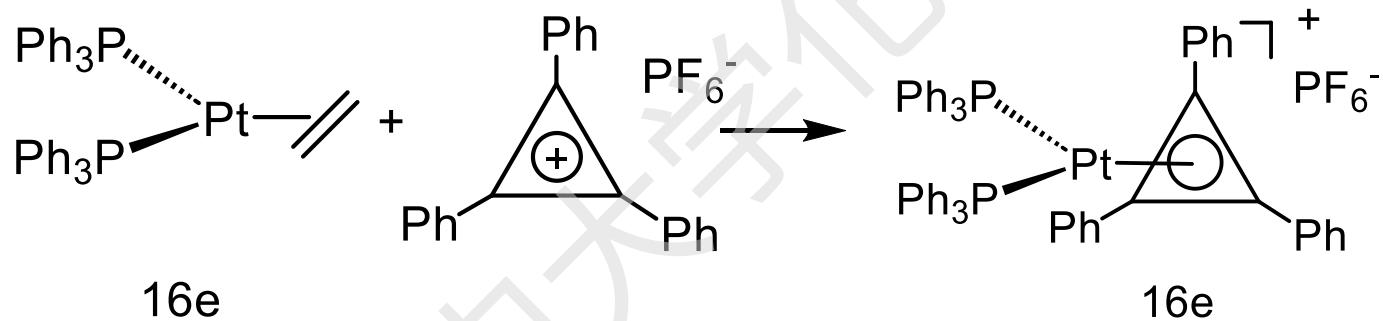
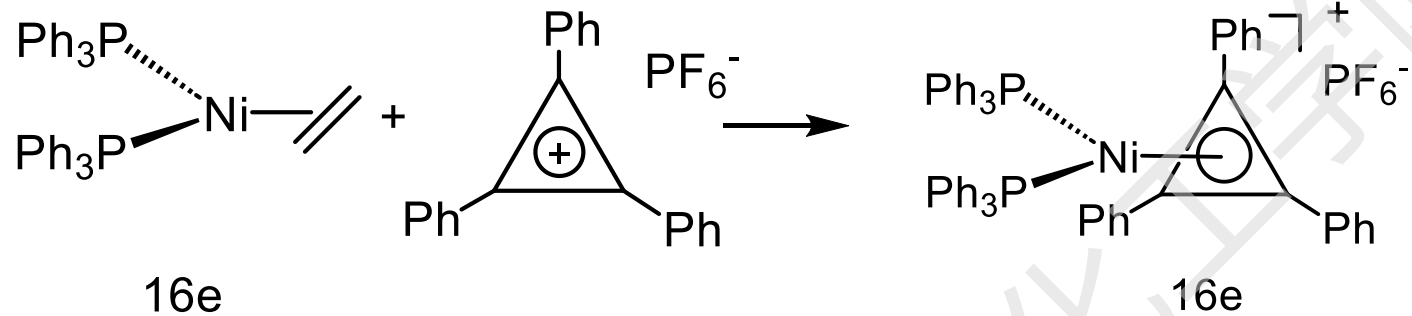


Examples and preparation



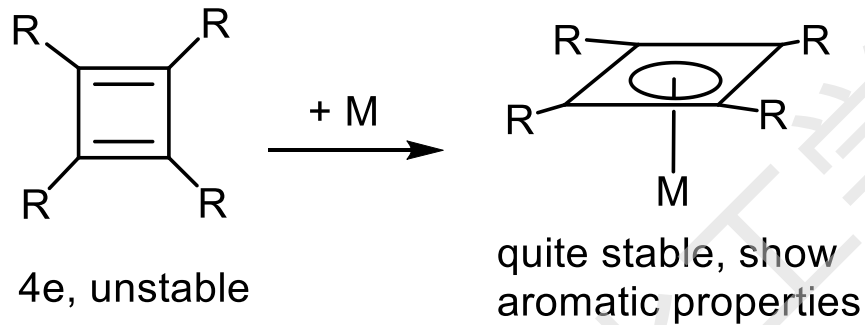


Examples and preparation

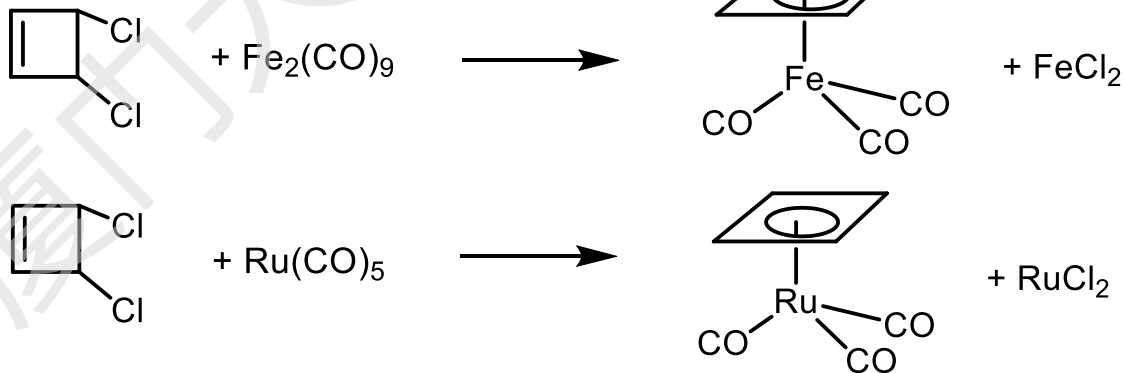
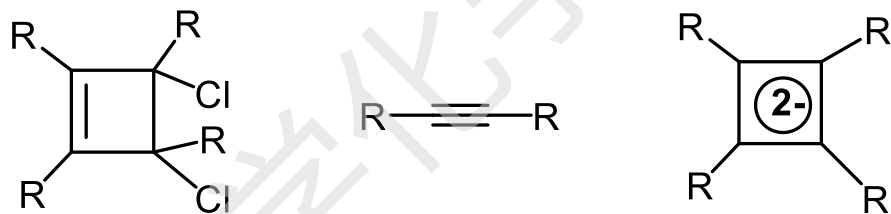




2) C_4R_4 as ligands

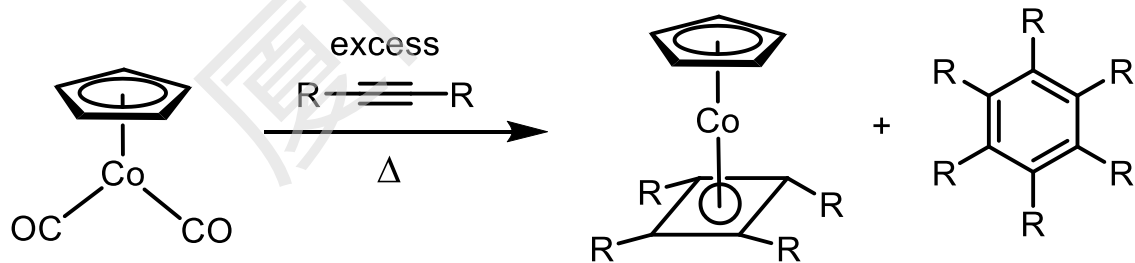
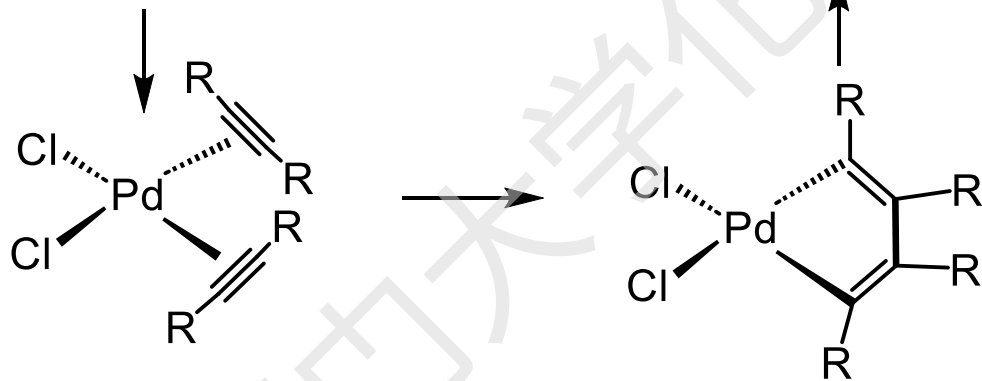
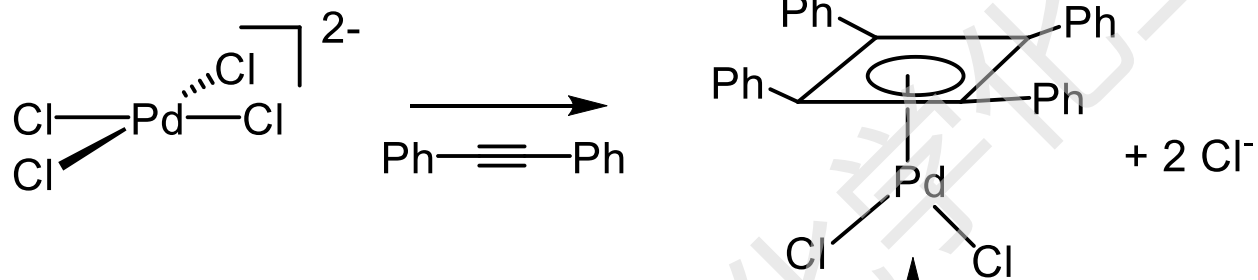
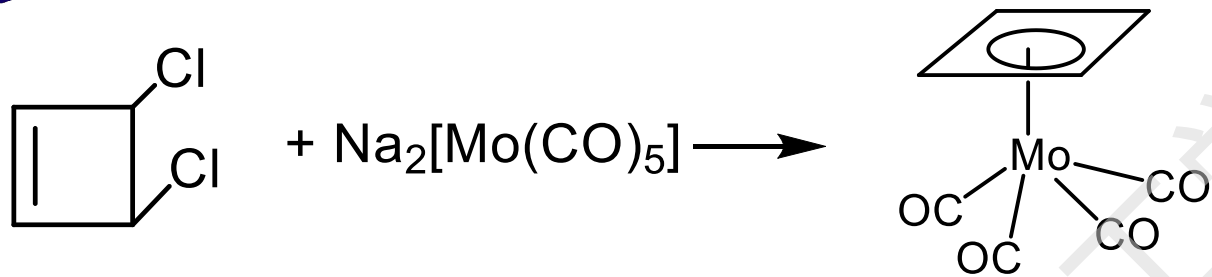


What are the common starting materials for preparing $M(\eta^4-C_4R_4)$ complexes?



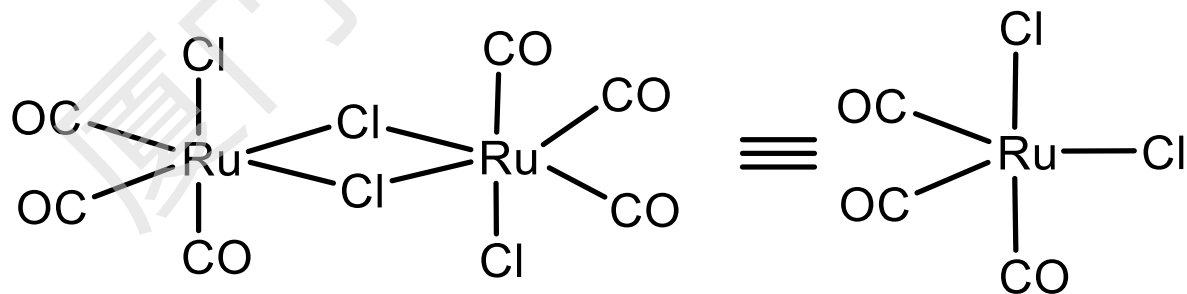
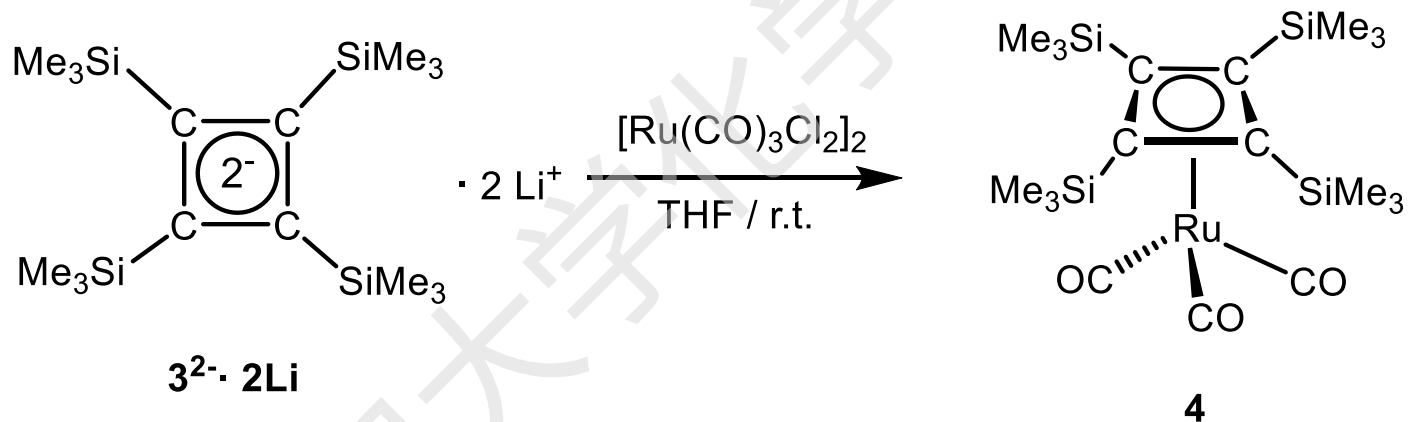
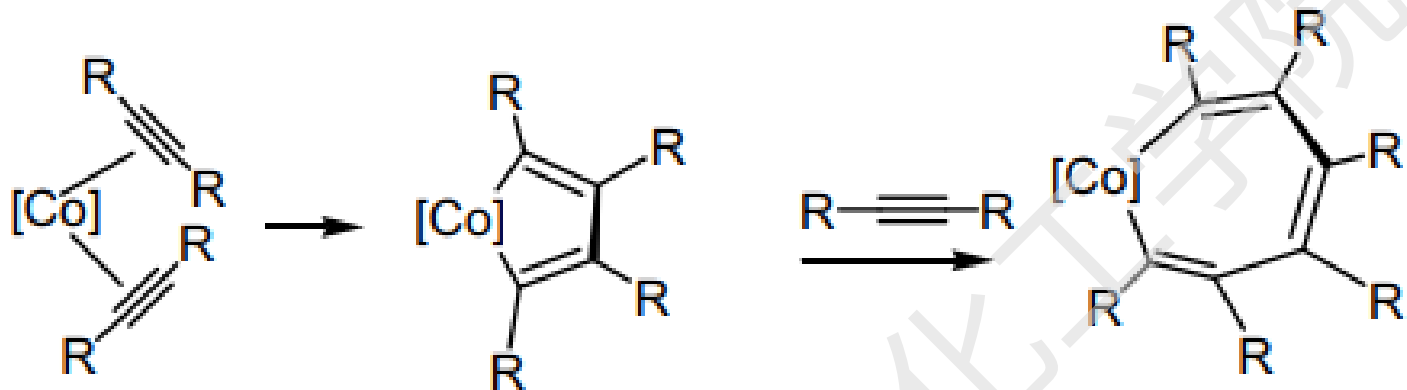


Example and preparation





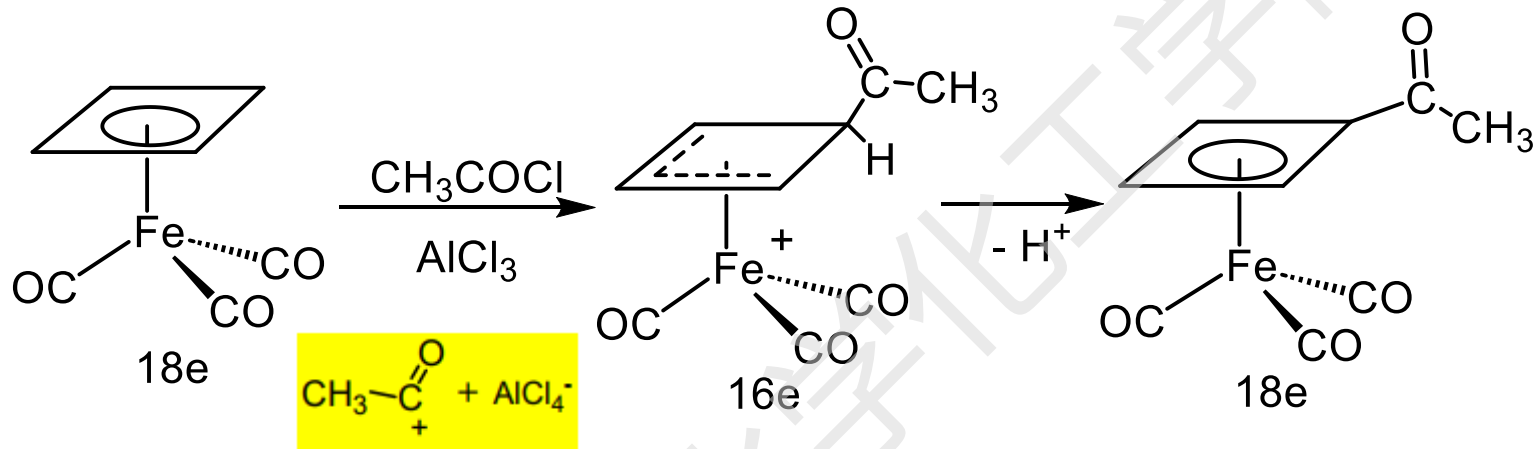
Example and preparation



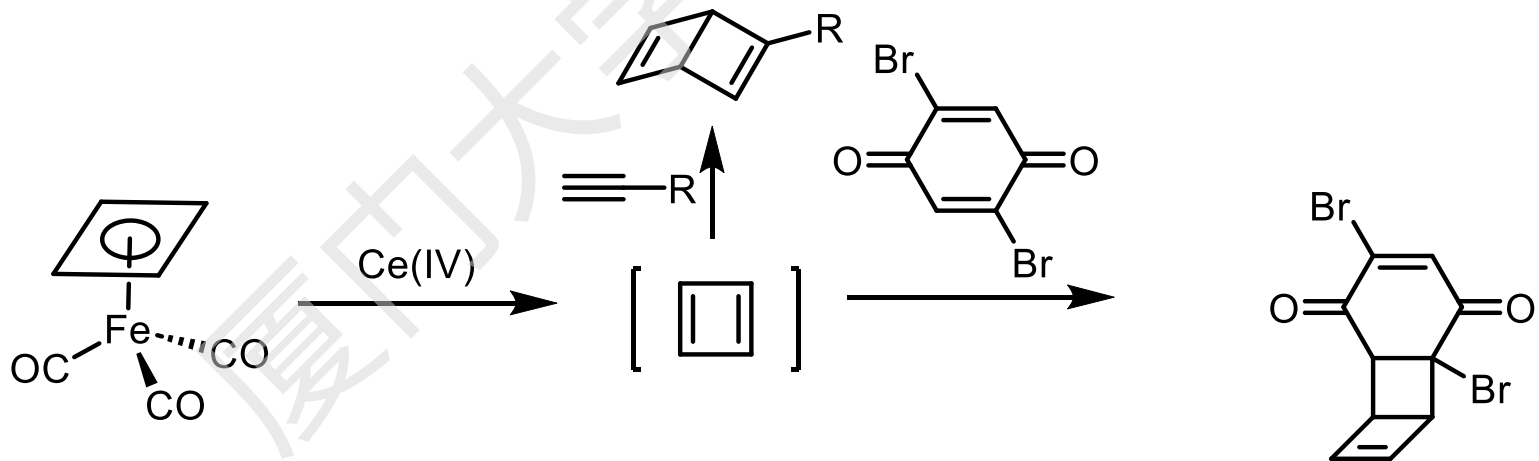


Important chemical properties

a. Show aromatic properties



b. Can be used as a source of C_4R_4

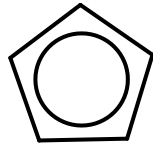




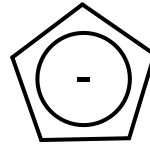
3) C_5R_5 as ligands

Common C_5R_5

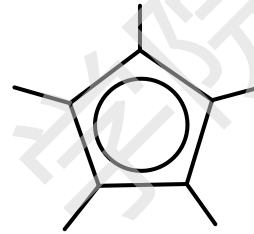
Cyclopentadienyl: Cp
Cyclopentadienide: Cp⁻



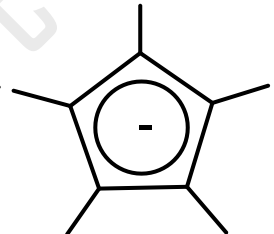
Cp
5e donor



Cp⁻
6e donor

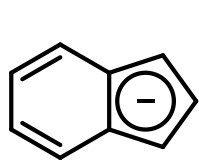
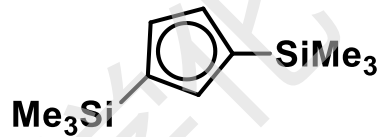
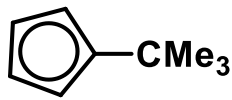
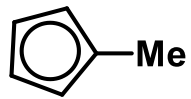


Cp*
5e donor

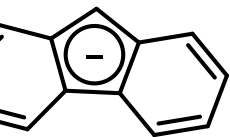
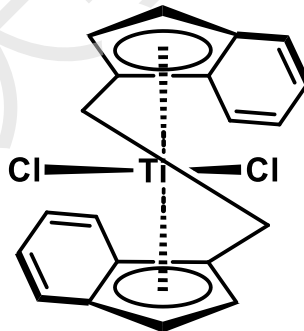
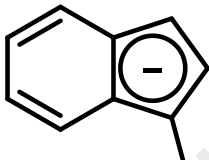


Cp*⁻
6e donor

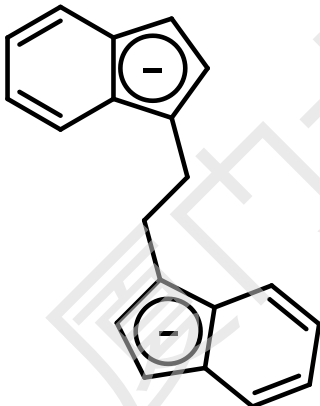
Other important Cp-type ligands:



Indenyl



Fluorenyl



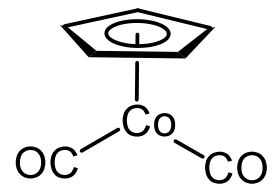
bis(indenyl)

Binding mode of C_5R_5

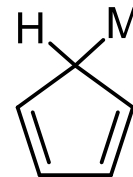
η^5 -Cp



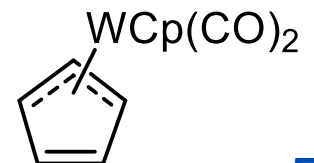
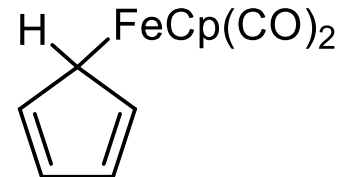
most common



η^1 -Cp

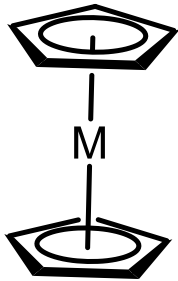


η^3 -Cp



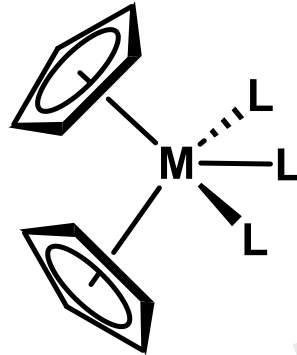


Common Types of Cp complexes:



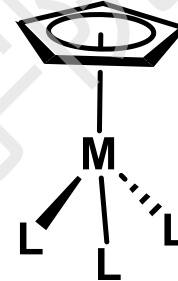
Metallocenes

Sandwich



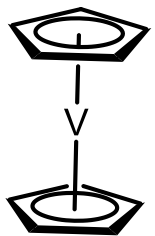
Bent Metallocenes

Tilted Sandwich



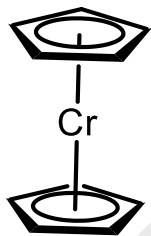
Half-Sandwich

Examples of metallocenes



d^3

15e



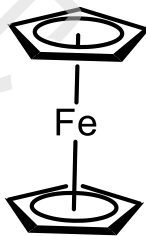
d^4

16e



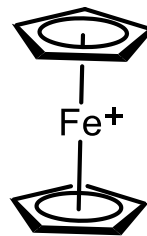
d^5

17e



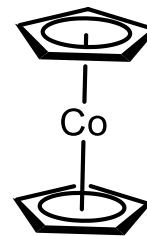
d^6

18e



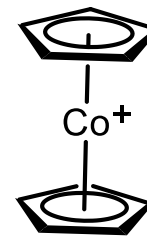
d^5

17e



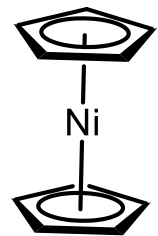
d^7

19e



d^6

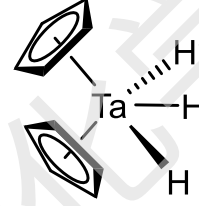
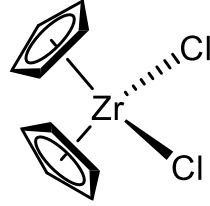
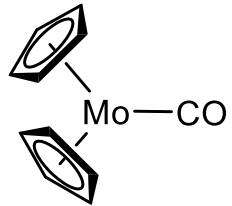
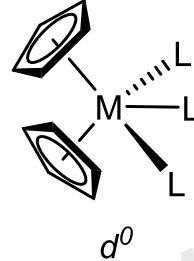
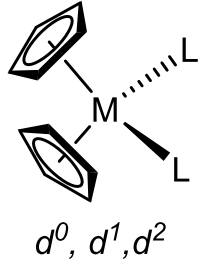
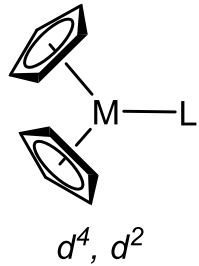
18e



d^8

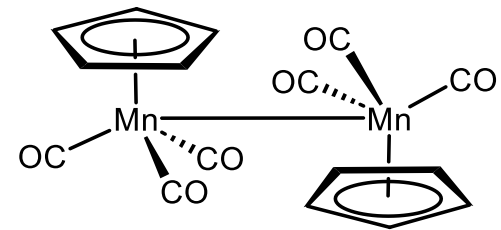
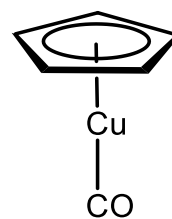
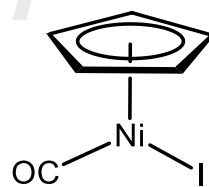
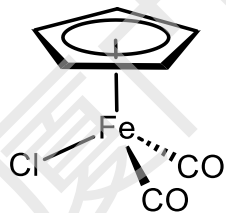
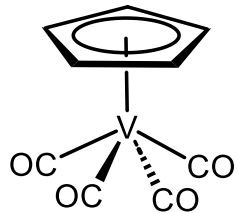
20e

Examples of Bent Metallocenes



Metals of group 4 and the 2nd and 3rd rows of groups 5-7 do not form stable Cp_2M complexes. However they do form stable metallocenes with additional ligands attached to the metal center.

Examples of Half-Sandwich Complexes





Examples of metallocenes

Properties of 3d metallocenes, MCp_2 .

FeCp_2	d^6 , 18-e	$R(\text{M-C})$ 2.06 Å
CoCp_2	d^7 , 19-e	2.12 Å
CrCp_2	d^4 , 16-e	2.17 Å
NiCp_2	d^8 , 20-e	2.20 Å
VCp_2	d^3 , 15-e	2.28 Å
MnCp_2	d^5 , 17-e	2.38 Å

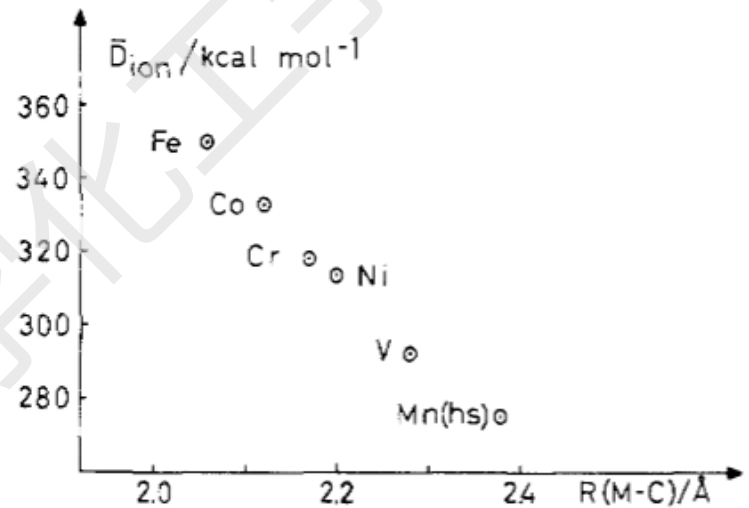
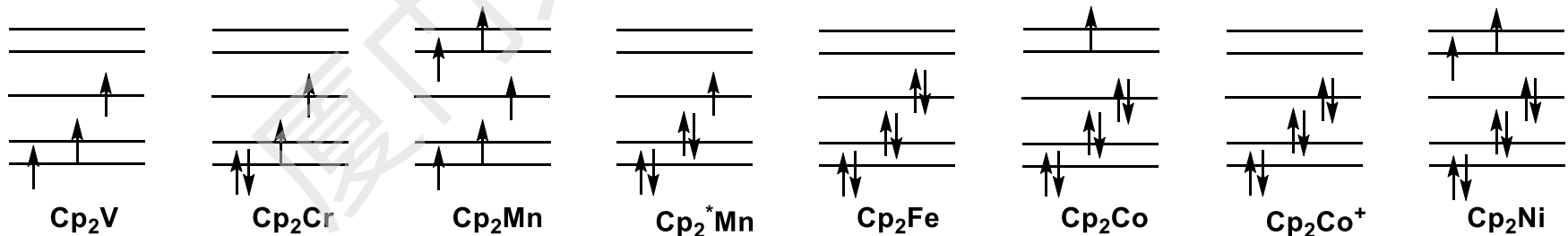


Figure 6. Mean ionic dissociation energy, $\bar{D}_{\text{ion}}(\text{M}^{2+}-\text{Cp}^-)$, of the 3d metallocenes as a function of the M-C bond distance.

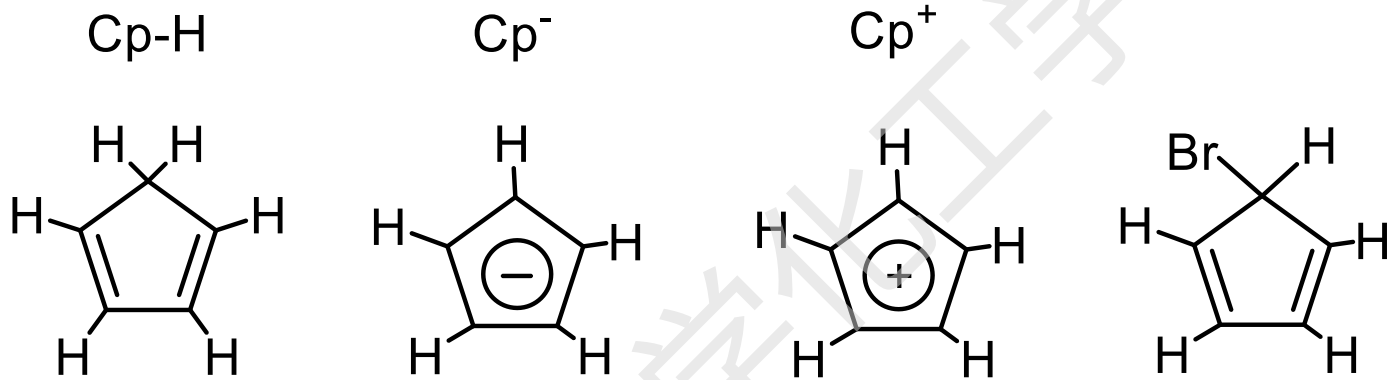
From Haaland, *A. Acc. Chem. Res.* **1979**, 415.





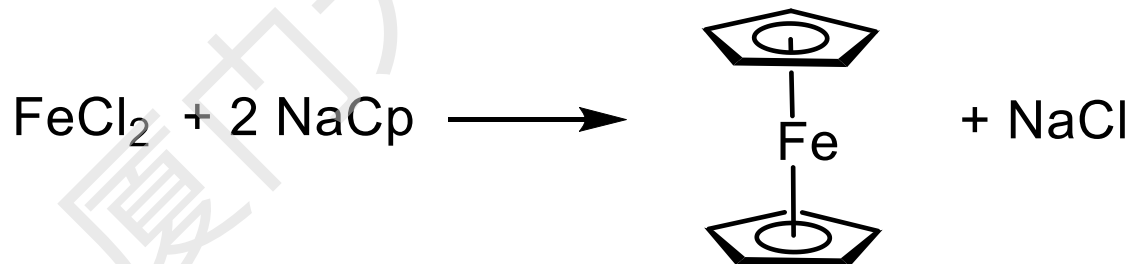
How to prepare CpM complexes?

Common starting materials:



i) From a source of Cp⁻ : most common route

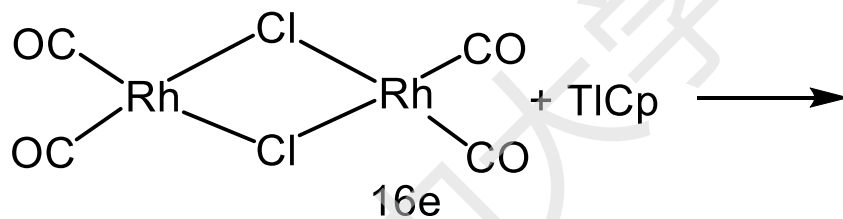
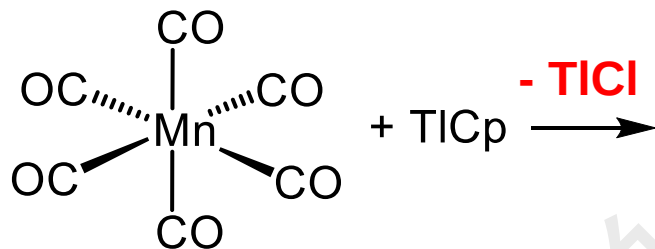
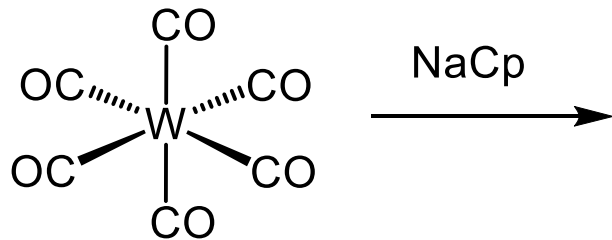
Cp⁻ reagents: NaCp, CpMgBr, TiCp, CpSnMe₃





Practice Problems

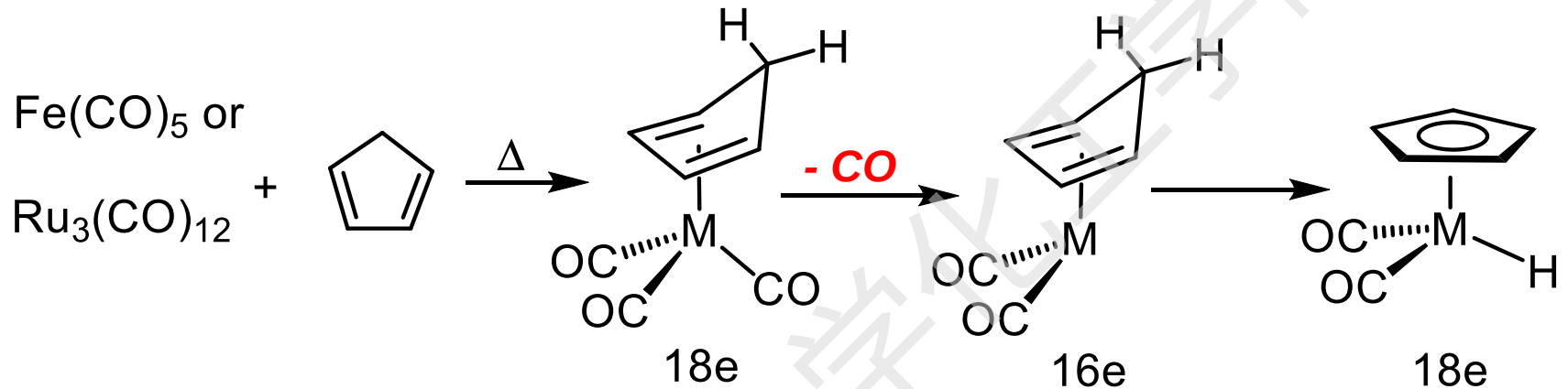
Predict the product of following reactions.



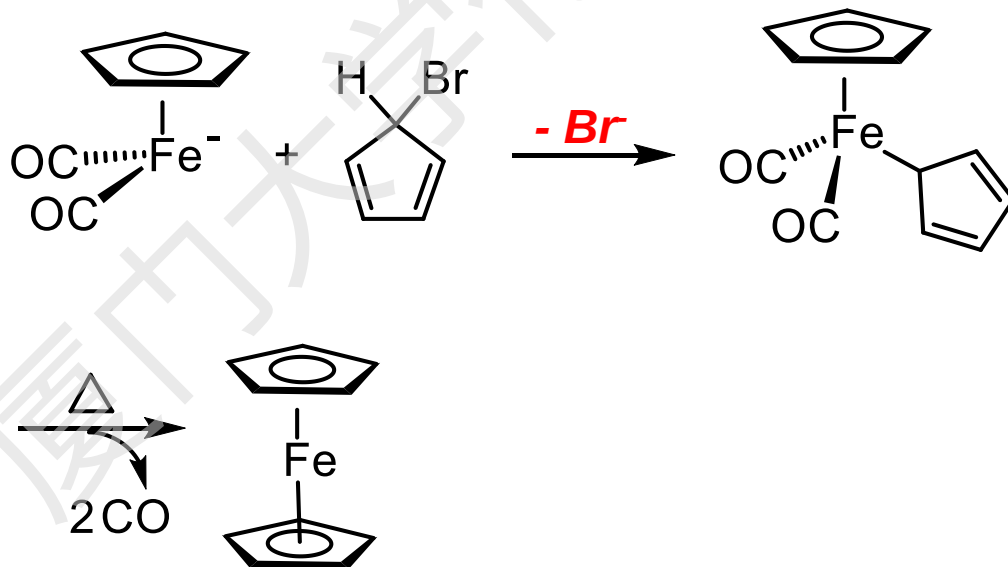


How to prepare CpM complexes?

ii) From C_5H_6 or related ligands.



iii) From a source of Cp^+ .



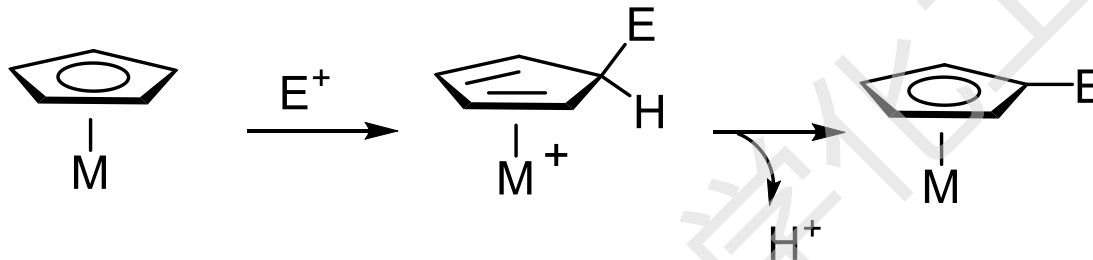


What are the typical chemical properties of $\eta^5\text{-Cp}$ complexes?

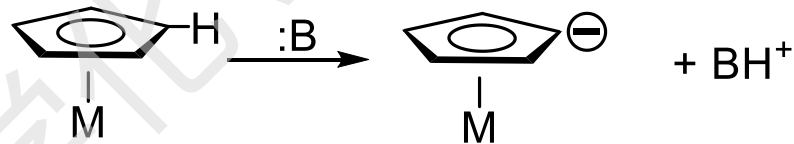
--- $\eta^5\text{-Cp}$ ligand is usually unreactive.

--- The following reactions may occur to $\eta^5\text{-Cp}$.

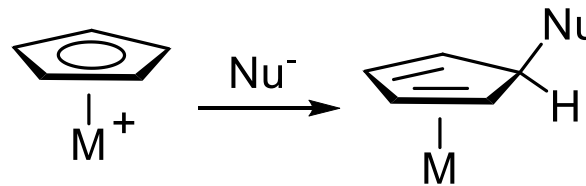
* Electrophilic substitution-like benzene



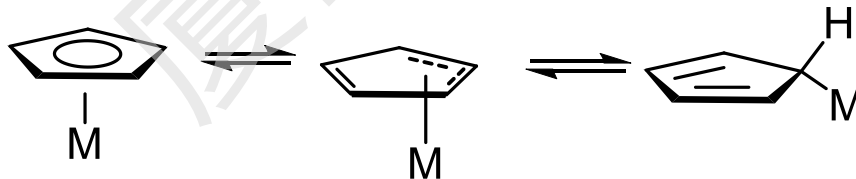
* Metallation reactions



* Nucleophilic attack on Cp

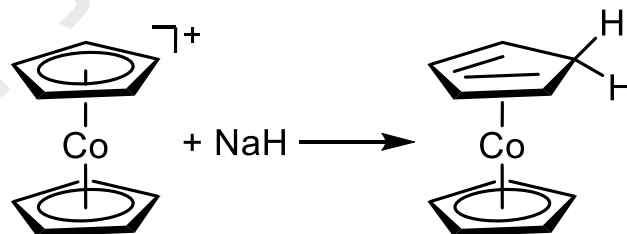
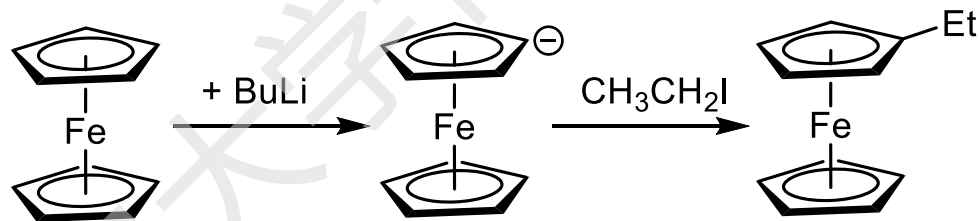
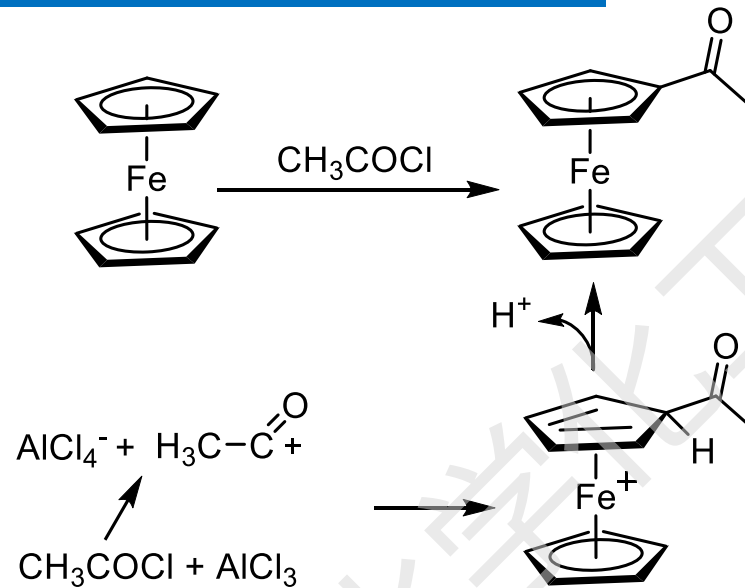


* $\eta^5\text{-Cp} \rightleftharpoons \eta^3\text{-Cp} \rightleftharpoons \eta^4\text{-Cp}$ interconversion





Chemical properties



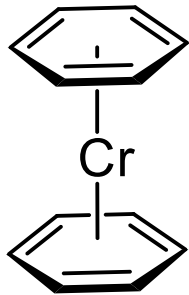


4) Arene complexes

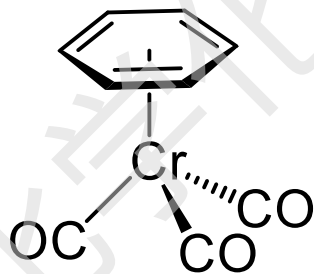
a) Common structures

■ As simple η^6 , η^4 , η^2 ligands.

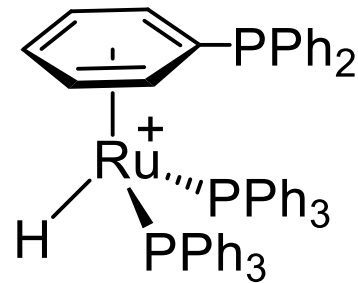
η^6 - most common



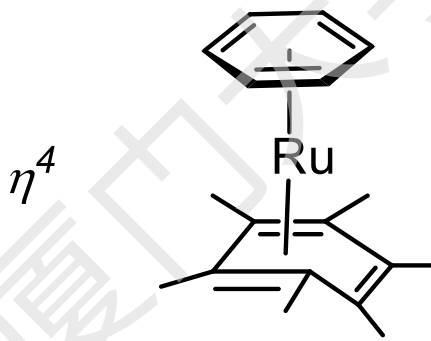
18e



18e

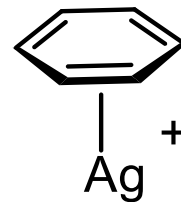


18e

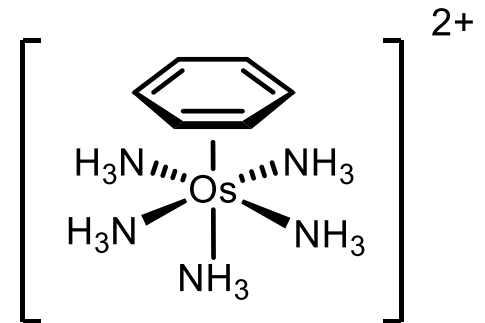


18e

η^2



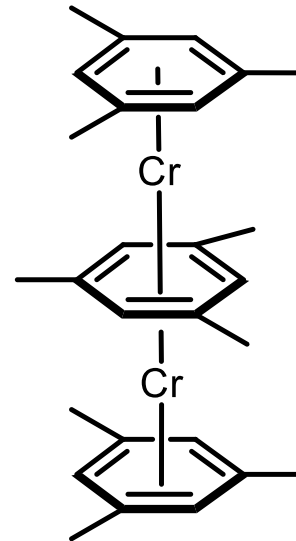
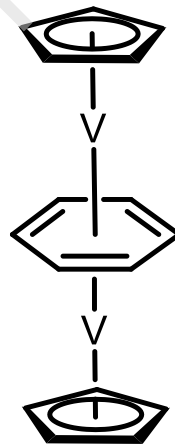
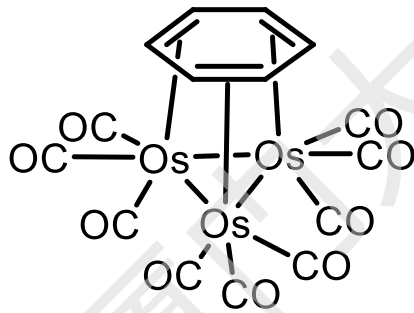
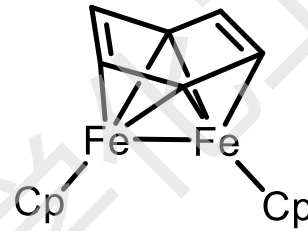
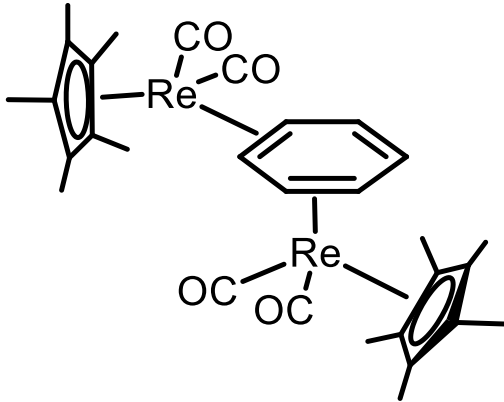
16e





4) Arene complexes

■ As bridging ligands (*for information*)





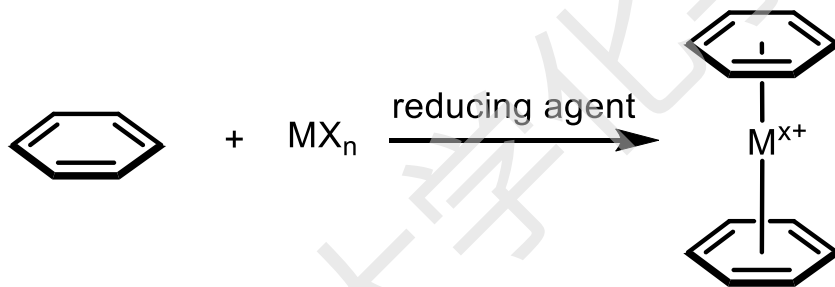
b) Common synthetic routes

1) From benzene

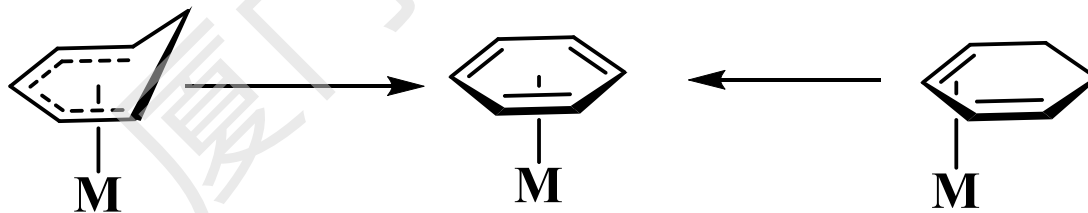
By substitution reactions



Benzene + metal salts + reducing agents

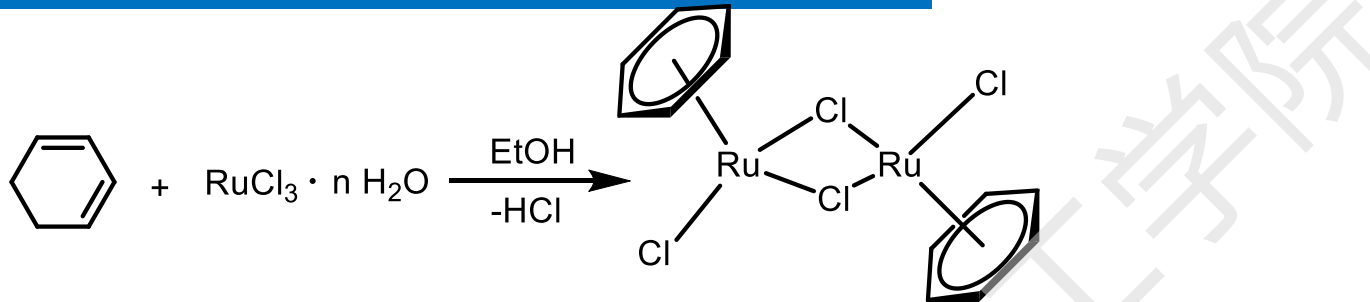


2) From reactions of coordinated ligands

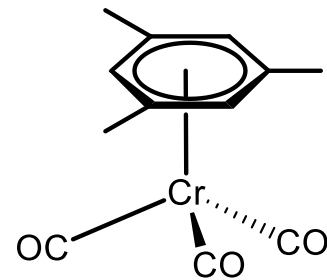
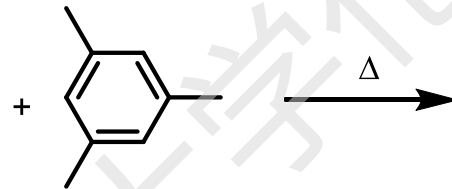
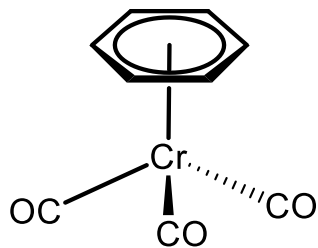
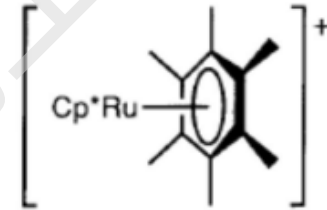
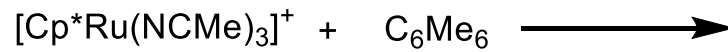




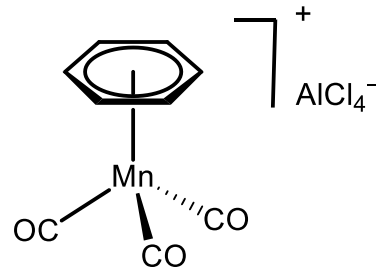
Additional examples



18 e⁻, Ru (II), d⁶



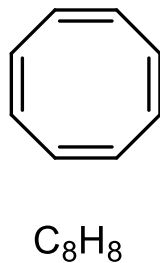
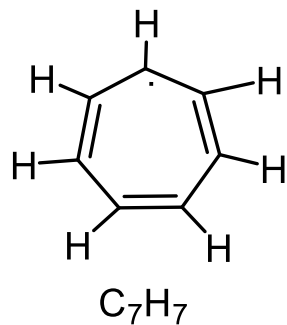
18 e⁻, Ru (II), d⁶



18 e⁻, Mn (I), d⁶

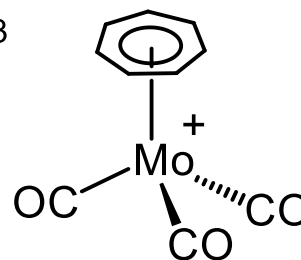
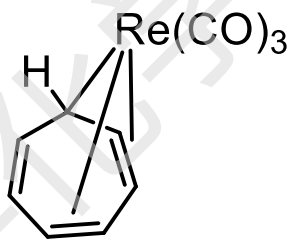
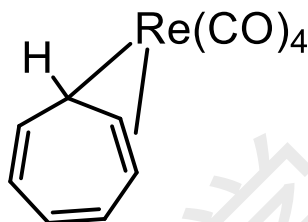
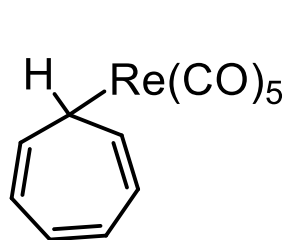


5) C_7H_7 and C_8H_8 as ligands

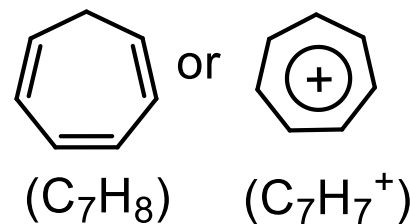


COT
cyclooctatetrene

• C_7H_7 can be a η^1 , η^3 , η^5 and η^7 ligand. e.g.



* C_7H_7 complexes can be made from

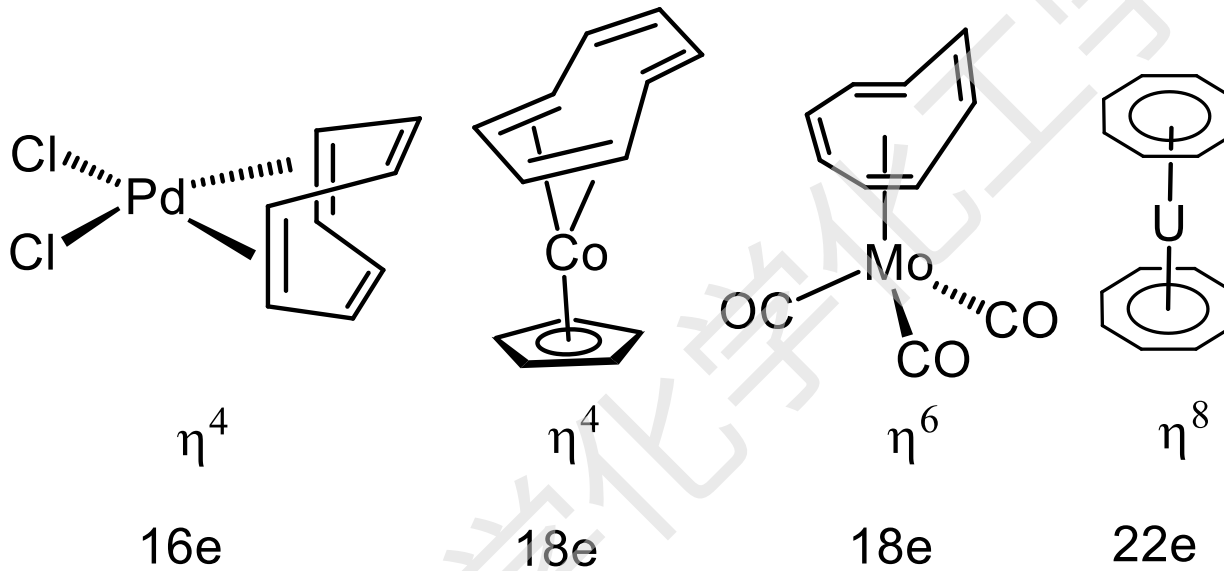


6e, aromatic

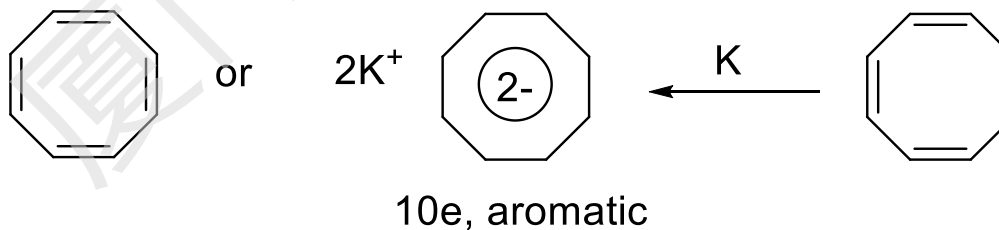


5) C_7H_7 and C_8H_8 as ligands

- C_8H_8 can be η^2 , η^4 , η^6 and η^8 ligands. *e.g.*



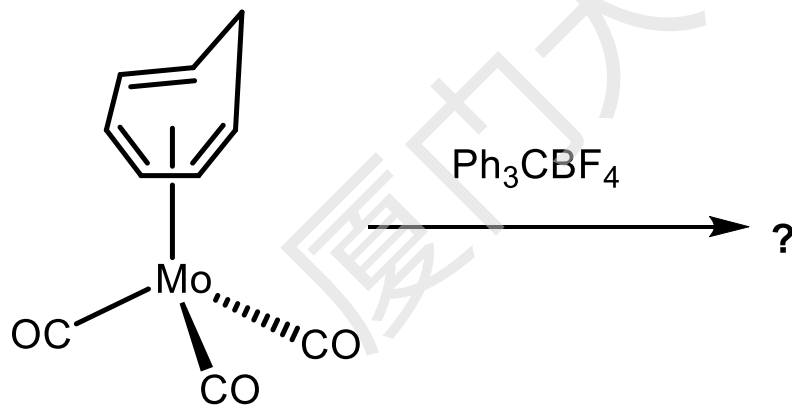
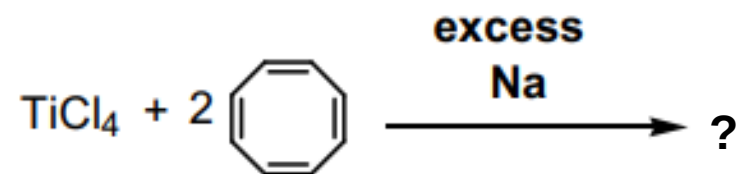
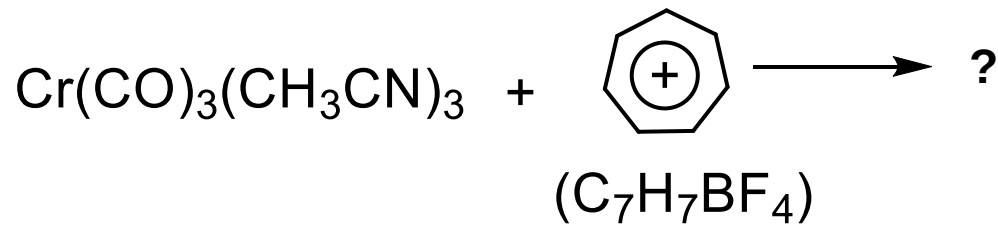
- COT complexes can be made from:





Practice Problems

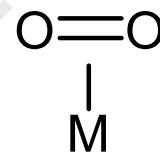
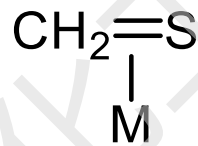
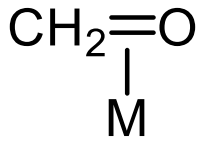
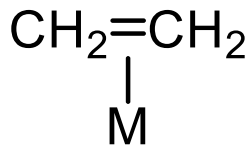
Suggest the products or reagents for the following reactions.





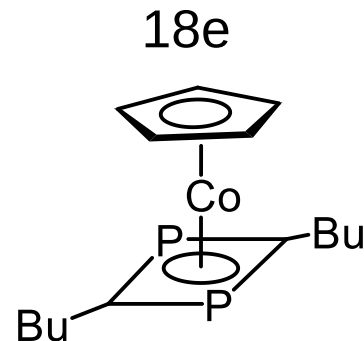
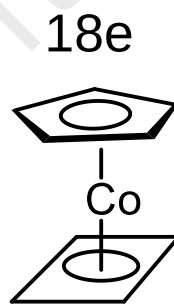
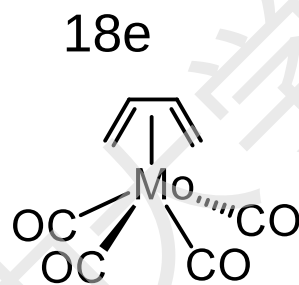
F. Metal π -complexes of unsaturated ligands with heteroatoms.

Many unsaturated ligands with heteroatoms can form complexes.



CH₂, O, and S are called isolobal

They have same number of valence electrons (6)



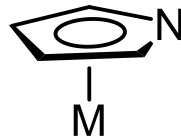
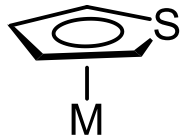
CH, N, and P are also isolobal (have 5 valence electrons)

Similar complexes can be obtained by replace a hydrocarblyl group in an organometallic compound with an isolobal group

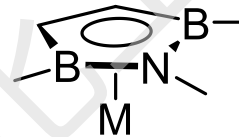
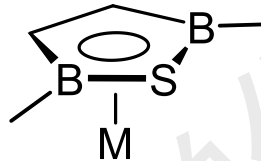
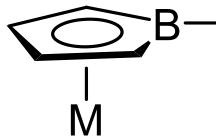


Other examples of heterocyclic π -complexes

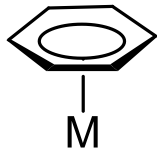
6e donor:



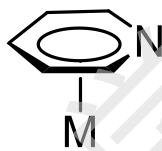
4e donor:



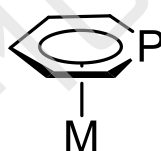
Donor: 6e



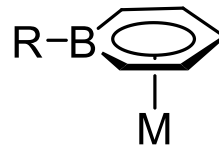
6e



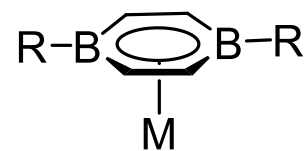
6e



5e

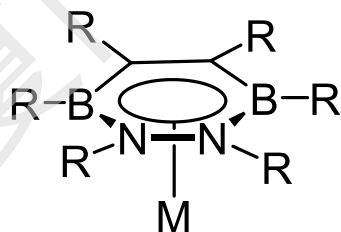


4e

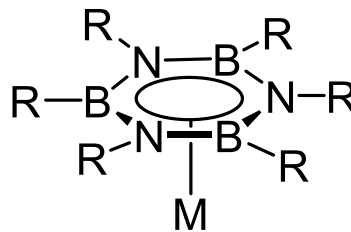


Donor:

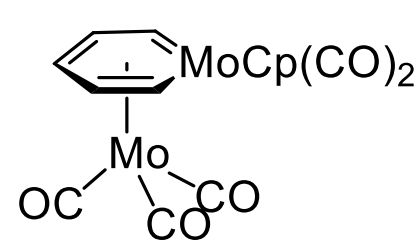
6e



6e



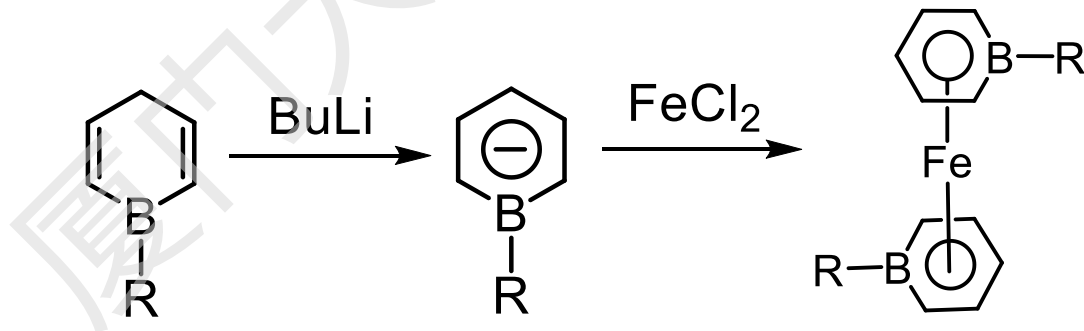
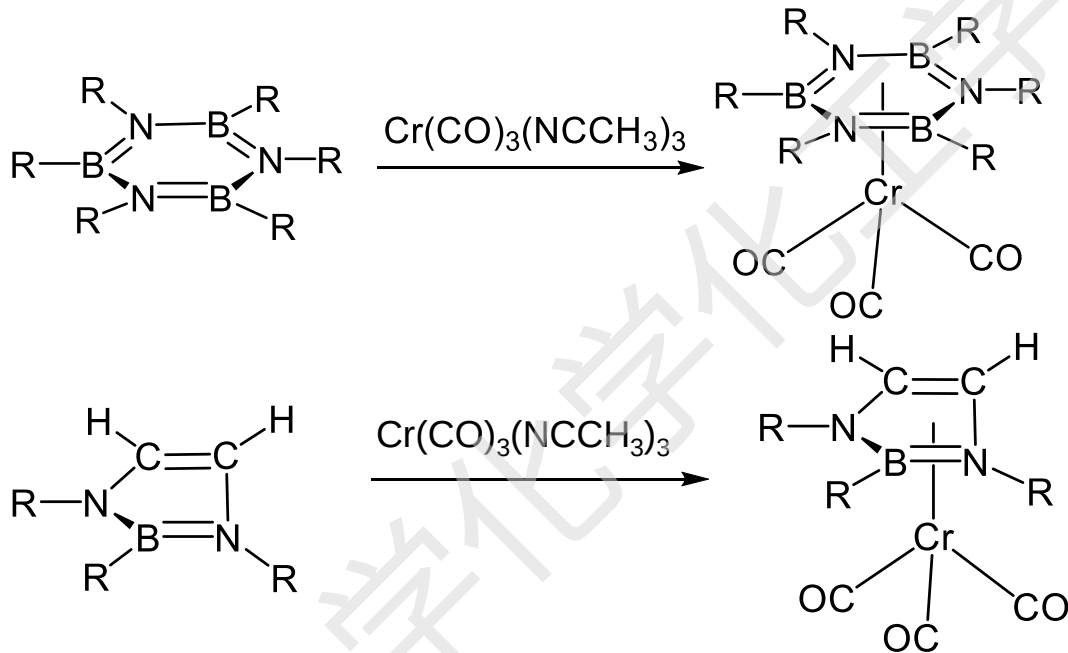
6e





Synthesis: similar to organic ones. e.g.

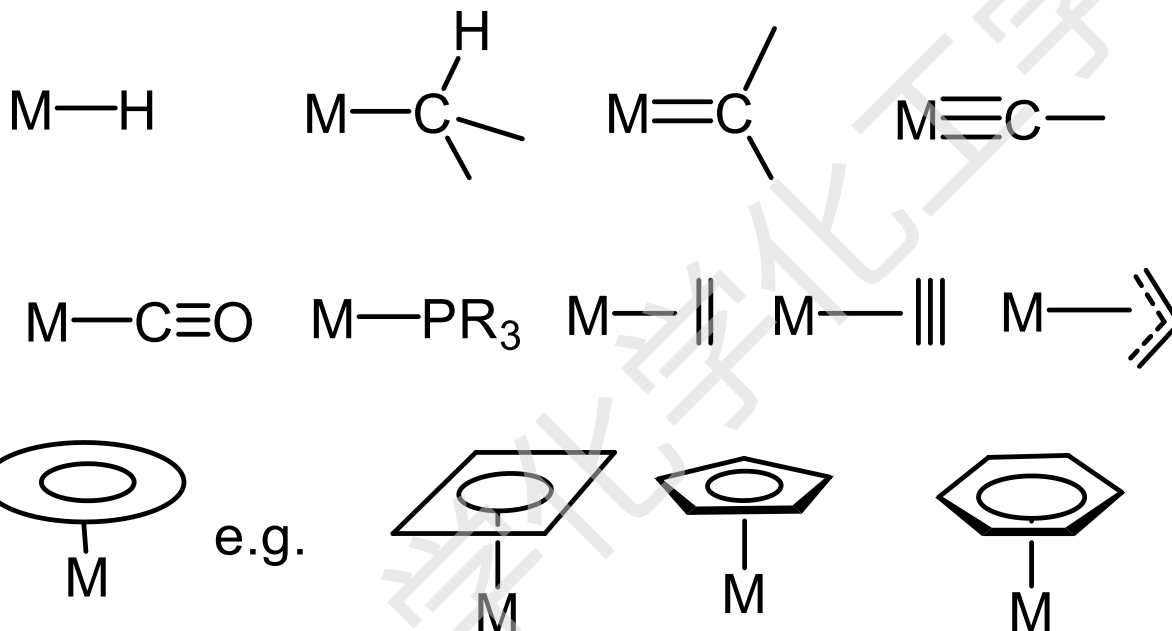
Give the reagents or products for the following preparations.





G. Characterization of Organometallic Compounds.

Common Organometallic Functionalities



How to characterize them?



Common spectroscopic methods (**for information**)

Chemical analysis: Composition

MS: Composition and functional groups

IR: Functional groups

NMR: Functional groups and stereochemistry

ESR: Paramagnetic compounds

X-ray: 3-D structures



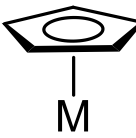
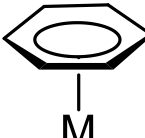
Common spectroscopic methods (for information)

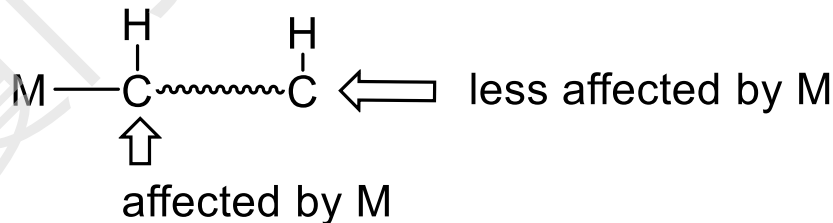
1) Characteristic IR frequencies of some common organometallic groups:

functional group	$M-H$	$M-C\equiv O$	$M-C \begin{array}{l} \nearrow O \\ \searrow R \end{array}$	$M-\begin{array}{ } \text{ } \end{array}$
frequency, cm^{-1}	1900-2250	1750-2100	1530-1680	1700-2000

2) NMR (1H , ^{13}C) in characterization of organometallics

Chemical shifts $\implies$ functional groups. e.g.

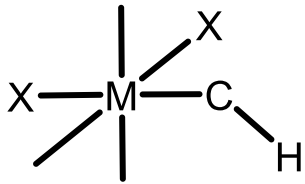
$M-H$	$M-\overset{\overset{H}{ }}{C}$	$M=\overset{\overset{H}{ }}{C}$	$M-\overset{\overset{H}{ }}{C}=\overset{\overset{O}{ }}{C}$		
$\delta(^1H) < 0$	~ 0	> 10	> 15	~ 5	~ 5
$\delta(^{13}C)$	~ 0	> 200	> 200	~ 90	~ 90





Common spectroscopic methods (for information)

Spin-spin coupling ==> stereochemistry



The signal of C(or H) may be split by X or M if they are also NMR active. e.g. when X is P, M is Rh or Pt.

$$J(\text{C-X})_{\text{cis}} < J(\text{C-X})_{\text{trans}} \quad J(\text{H-X})_{\text{cis}} < J(\text{H-X})_{\text{trans}}$$

Coupling	trans (Hz)	cis (Hz)
$^2J_{\text{H-P}}$	90-160 Hz	10-30 Hz
$^2J_{\text{P-P}}$	150-500 Hz	20-50 Hz



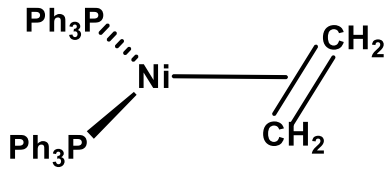
Common spectroscopic methods (for information)

Nucleus	Spin	Natural Abundance (%)	Frequency $^1\text{H} = 100 \text{ MHz}$	Receptivity $^{13}\text{C} = 1$
Ligands				
^1H	1/2	99.99	100	5870
^2H (D)	1	0.01	15.35	0.006
^{13}C	1/2	1.1	25.15	1.00
^{11}B	3/2	80	32.1	777
^{19}F	1/2	100	94.1	4900
^{31}P	1/2	100	40.5	391
Metals				
^{103}Rh	1/2	100	4.9	0.68
^{109}Ag	1/2	51.8	4.05	0.20
^{107}Ag	1/2	48.2	4.65	0.28
^{195}Pt	1/2	33.8	21.4	20.1

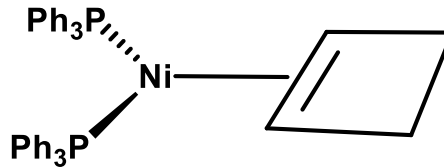


HOMework

1. Which of the following olefin complexes have a stronger Ni- olefin bond?

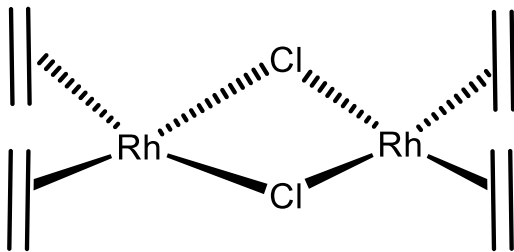


(a)

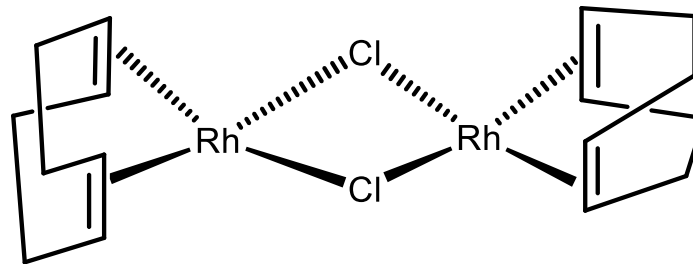


(b)

2. Which of the following olefin complexes is more stable?



(a)

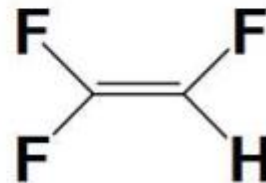
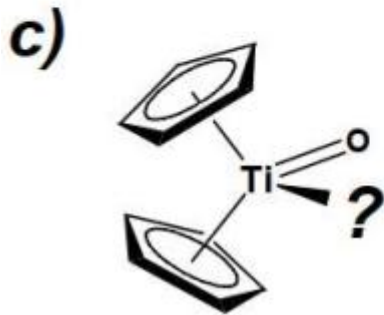
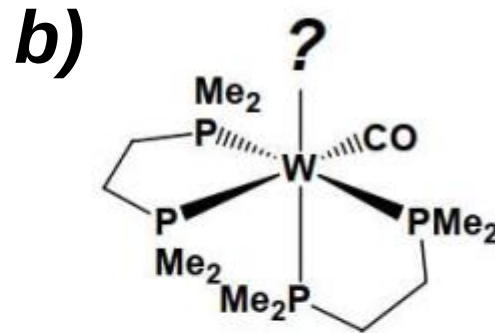
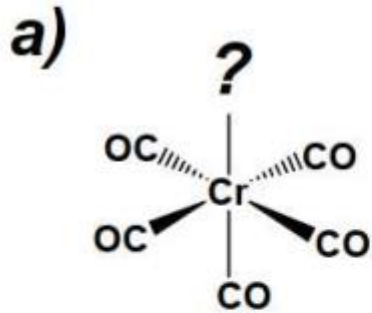


(b)



HOMework

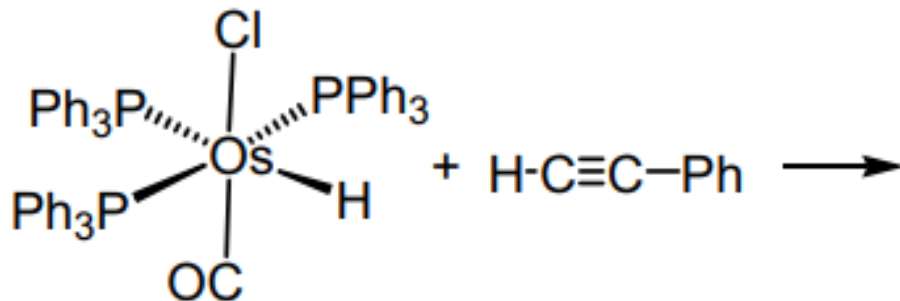
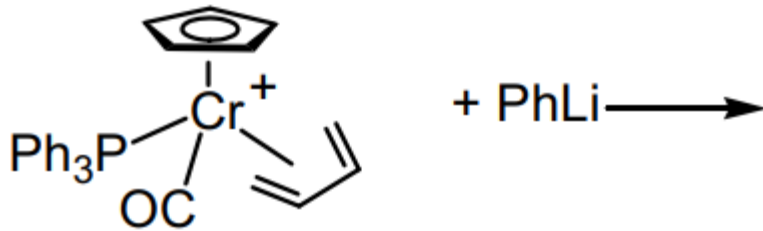
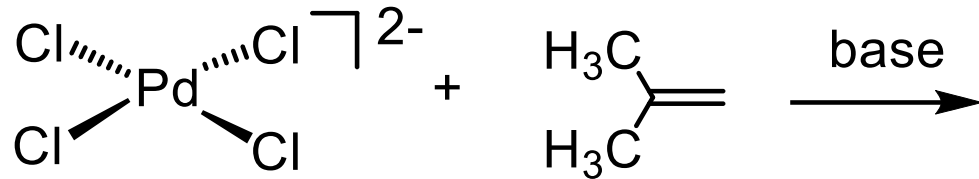
3. To which of the following (each with a single open coordination site) will trifluoroethylene bond to the most strongly





HOMework

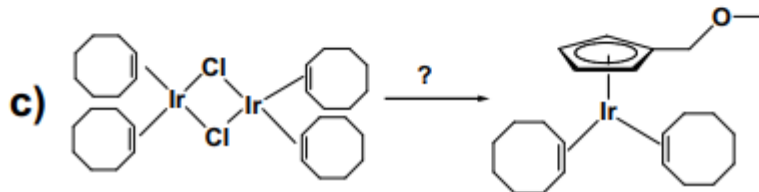
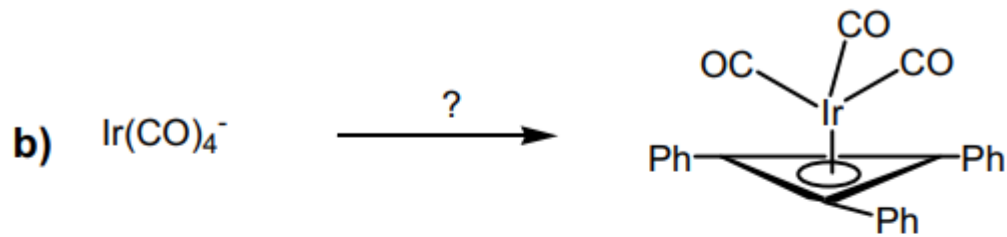
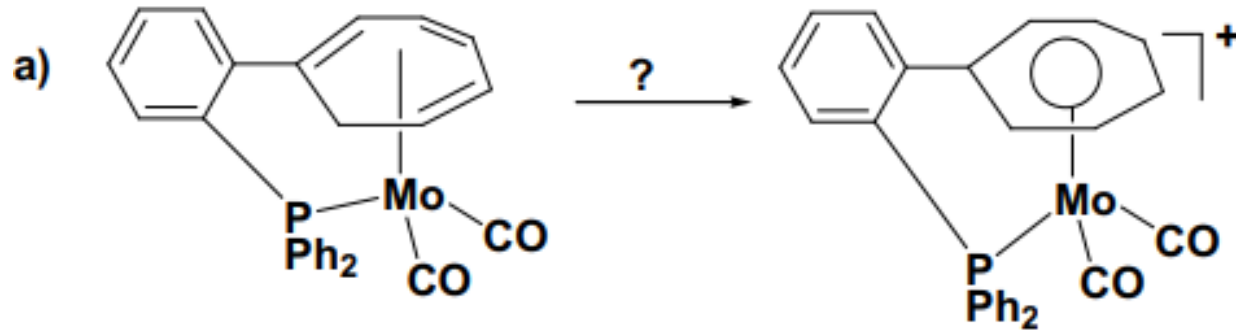
4. Predict the metal-containing products of the following reactions





HOMework

5. Provide appropriate reagents for the following transformations





HOMEWORK

6. Propose a reasonable mechanism for the following reaction

